

PART A

常微分方程式

Ordinary Differential Equations (ODE)

CHAPTER 1 First-Order ODEs

CHAPTER 2 Second-Order Linear ODEs

CHAPTER 3 Higher Order Linear ODEs

CHAPTER 4 Systems of ODEs. Phase Plane. Qualitative Methods

~~CHAPTER 5 Series Solutions of ODEs. Special Functions~~

CHAPTER 6 Laplace Transforms

許多物理定律與關係可用ODE表示，如牛頓第二運動定律 $F=ma=ma$

ODE：常微分方程式。
 y 為未知函數，其只有一個變數。

$$y = y(x)$$
$$y' = \frac{dy}{dx} \quad y'' = \frac{d^2y}{dx^2}$$

PDE：偏微分方程式。
 y 為未知函數，其有兩個(以上)變數。

$$y = y(x, t)$$

$$y'_x = \frac{\partial y}{\partial x}$$

$$y'_t = \frac{\partial y}{\partial t}$$

$$y''_{xt} = \frac{\partial^2 y}{\partial t \partial x}$$

先對 x 微分，再對 t 微分

Many physical laws and relations can be expressed mathematically in the form of differential equations. Thus it is natural that this book opens with the study of differential equations and their solutions. Indeed, many engineering problems appear as differential equations.

The main objectives of Part A are twofold: the study of ordinary differential equations and their most important methods for solving them and the study of modeling.

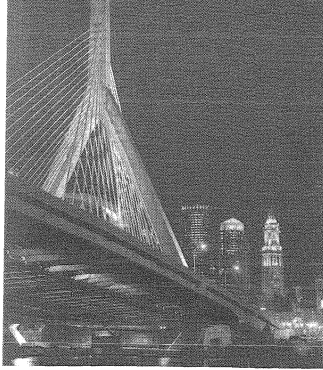
Ordinary differential equations (ODEs) are differential equations that depend on a single variable. The more difficult study of partial differential equations (PDEs), that is, differential equations that depend on several variables, is covered in Part C.

建模：物理實體→數學方程式

Modeling is a crucial general process in engineering, physics, computer science, biology, medicine, environmental science, chemistry, economics, and other fields that translates a physical situation or some other observations into a "mathematical model." Numerous examples from engineering (e.g., mixing problem), physics (e.g., Newton's law of cooling), biology (e.g., Gompertz model), chemistry (e.g., radiocarbon dating), environmental science (e.g., population control), etc. shall be given, whereby this process is explained in detail, that is, how to set up the problems correctly in terms of differential equations.

For those interested in solving ODEs numerically on the computer, look at Secs. 21.1–21.3 of Chapter 21 of Part F, that is, numeric methods for ODEs. These sections are kept independent by design of the other sections on numerics. This allows for the study of numerics for ODEs directly after Chap. 1 or 2.

以電腦程式求ODE之解→數值方法求近似解



CHAPTER 1

First-Order ODEs

$$y' = \cos x$$

$$xy' = 2y$$

Chapter 1 begins the study of ordinary differential equations (ODEs) by deriving them from physical or other problems (*modeling*), solving them by standard mathematical methods, and interpreting solutions and their graphs in terms of a given problem. The simplest ODEs to be discussed are *ODEs of the first order* because they involve only the first derivative of the unknown function and no higher derivatives. These unknown functions will usually be denoted by $y(x)$ or $y(t)$ when the independent variable denotes time t . The chapter ends with a study of the existence and uniqueness of solutions of ODEs in Sec. 1.7.

Understanding the basics of ODEs requires solving problems by hand (paper and pencil, or typing on your computer, but first without the aid of a CAS). In doing so, you will gain an important conceptual understanding and feel for the basic terms, such as ODEs, direction field, and initial value problem. If you wish, you can use your **Computer Algebra System (CAS)** for checking solutions.

COMMENT. *Numerics for first-order ODEs can be studied immediately after this chapter.* See Secs. 21.1–21.2, which are independent of other sections on numerics.

Prerequisite: Integral calculus.

Sections that may be omitted in a shorter course: 1.6, 1.7.

學習重點：

1.建模的概念；2.認識ODE與PDE；3.解的觀念；4.通解與特解的差異；5.初始值問題；6.解題(求解)步驟的認識

本課程不會執行數值方法，但未來研究常會使用數值法求解，故寫程式的功力很重要！！

1.1 Basic Concepts. Modeling

建模：物理實體→數學方程式→求解→解釋現象

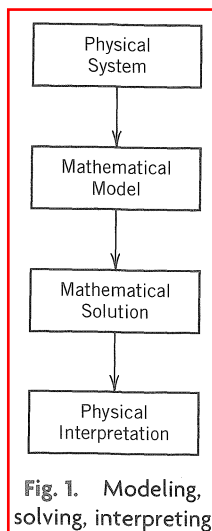


Fig. 1. Modeling, solving, interpreting

If we want to solve an engineering problem (usually of a physical nature), we first have to formulate the problem as a mathematical expression in terms of variables, functions, and equations. Such an expression is known as a mathematical model of the given problem. The process of setting up a model, solving it mathematically, and interpreting the result in physical or other terms is called *mathematical modeling* or, briefly, **modeling**. 解工程(或物理)問題，先要學會如何Modeling

Modeling needs experience, which we shall gain by discussing various examples and problems. (Your computer may often help you in *solving* but rarely in *setting up* models.)

Now many physical concepts, such as velocity and acceleration, are derivatives. Hence a model is very often an equation containing derivatives of an unknown function. Such a model is called a **differential equation**. Of course, we then want to find a solution (a function that satisfies the equation), explore its properties, graph it, find values of it, and interpret it in physical terms so that we can understand the behavior of the physical system in our given problem. However, before we can turn to methods of solution, we must first define some basic concepts needed throughout this chapter.

工程數學問題的建模與分析步驟

(一) 觀察與確認物理系統

1. 固力學問題
2. 振動學問題
3. 熱傳學問題
4. 流力學問題
5. 機構學問題
6. 電子電路問題
-等等工程領域問題。

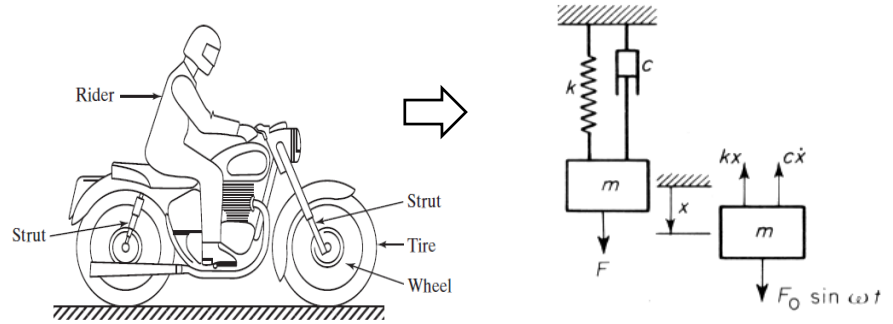


圖 1. 簡諧振動系統示意圖

不同物理系統，必須建立不同類型的數學方程式，因此必須有不同的解方程式技巧。

(二) 建立系統數學方程式(Governing equation) (建模)

建模就是將實際的系統以一具有相同特性的物理模型來取代，並將物理模型依定律、法則描述成數學模式。透過假設把實際的系統行為以數學語言表達出來，特別是在電腦普及便利的今天，利用數學模型有利於分析、設計與模擬。例如設計衛星的姿態控制器，我們不可能把一枚衛星直接搬到設計桌上設計控制器，亦不可能每設計一控制器就直接施加於衛星上作測試。

例：圖 1 系統之運動方程式 $m\ddot{x} + c\dot{x} + kx = F_0 \sin \omega t$

$\Sigma F = m a$ (向下為正)

求解 x (工數技巧)

(三) 工程數學之應用(數學方程式求解)

1. 常微分方程式
2. Laplace 轉換
3. 線性代數：向量、矩陣、行列式
4. 向量微分與向量積分
5. Fourier 級數積分與轉換
6. 偏微分方程式
-等等解題技巧。

解常微分方程式得

$$x = X \sin(\omega t - \phi) \quad , \quad \text{其中}$$

$$X = \frac{F_0}{\sqrt{(k - m\omega^2)^2 + (c\omega)^2}}$$

振幅

$$\phi = \tan^{-1} \frac{c\omega}{k - m\omega^2}$$

相位

工程數學即用以解不同類型數學方程式(理工領域)之一門學問，不同方程式需要不同解題技巧。

(四) 解釋結果與工程應用
寫程式run出數值，
並電腦繪出圖形

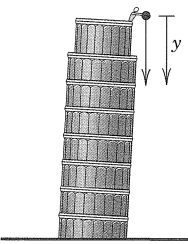
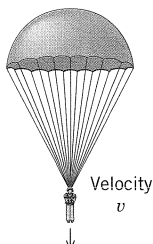
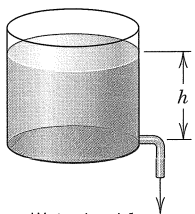
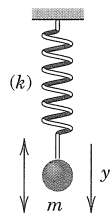
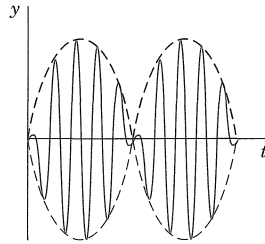
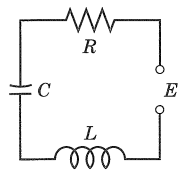
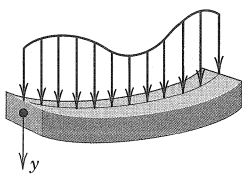
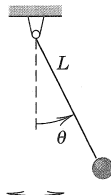
 Falling stone $y'' = g = \text{const.}$ (Sec. 1.1)	 Parachutist $mv' = mg - bv^2$ (Sec. 1.2)	 Water level h Outflowing water $h' = -k\sqrt{h}$ (Sec. 1.3)
 Displacement y Vibrating mass on a spring $my'' + ky = 0$ (Secs. 2.4, 2.8)	 Beats of a vibrating system $y'' + \omega_0^2 y = \cos \omega t, \quad \omega_0 \approx \omega$ (Sec. 2.8)	 Current I in an RLC circuit $LI'' + RI' + \frac{1}{C}I = E'$ (Sec. 2.9)
 Deformation of a beam $EIy^{iv} = f(x)$ (Sec. 3.3)	 Pendulum $L\theta'' + g \sin \theta = 0$ (Sec. 4.5)	 掠食者之餌食模型 Lotka-Volterra predator-prey model $\begin{aligned} y_1' &= \alpha y_1 - \beta y_1 y_2 \\ y_2' &= \beta y_1 y_2 - \gamma y_2 \end{aligned}$ (Sec. 4.5)

Fig. 2. Some applications of differential equations

本課程將著重在機電工程問題。

An **ordinary differential equation (ODE)** is an equation that contains one or several derivatives of an unknown function, which we usually call $y(x)$ (or sometimes $y(t)$ if the independent variable is time t). The equation may also contain y itself, known functions of x (or t), and constants. For example,

(1) 1階ODE

$$y' = \cos x$$

(2) 2階ODE

$$y'' + 9y = e^{-2x}$$

(3) 3階ODE

$$y'(y''') - \frac{3}{2}y'^2 = 0$$

愈高階計算越複雜

只有一個變數

常微分 $y = y(x) \Rightarrow y' = \frac{dy}{dx} \Rightarrow y'' = \frac{d^2y}{dx^2}$

$Dy = y'$
 $D^2y = y''$
 $D^3y = y'''$
 D : 微分運算子

$3y''' + 5y'' + 2y' + 4y = 5$
 $\Rightarrow 3D^3y + 5D^2y + 2Dy + 4y = 5$
 $\Rightarrow (3D^3 + 5D^2 + 2D + 4I)y = 5$

are ordinary differential equations (ODEs). Here, as in calculus, y' denotes dy/dx , $y'' = d^2y/dx^2$, etc. The term *ordinary* distinguishes them from *partial differential equations* (PDEs), which involve partial derivatives of an unknown function of two or more variables. For instance, a PDE with unknown function u of two variables x

$u = u(x, y)$

$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0.$

PDE (偏微分方程式)

偏微分 $y = y(x, t) \Rightarrow y'_x = \frac{\partial y}{\partial x} \Rightarrow y''_{xx} = \frac{\partial^2 y}{\partial x^2}$

$\Rightarrow y'_t = \frac{\partial y}{\partial t} \Rightarrow y''_{xt} = \frac{\partial^2 y}{\partial t \partial x}$ (先對 x 微分, 再對 t 微分)

PDEs have important engineering applications, but they are more complicated than ODEs; they will be considered in Chap. 12. **階數**

An ODE is said to be of **order n** if the n th derivative of the unknown function y is the highest derivative of y in the equation. The concept of order gives a useful classification into ODEs of first order, second order, and so on. Thus, (1) is of first order, (2) of second order, and (3) of third order.

In this chapter we shall consider **first-order ODEs**. Such equations contain only the first derivative y' and may contain y and any given functions of x . Hence we can write them as

一階ODE的型式

(4)

or often in the form

外在形式

$F(x, y, y') = 0$

內在形式

$y' = f(x, y).$

This is called the **explicit form**, in contrast to the **implicit form** (4). For instance, the implicit ODE $x^{-3}y' - 4y^2 = 0$ (where $x \neq 0$) can be written explicitly as $y' = 4x^3y^2$.

$x^{-3}y' - 4y^2 = 0 \Rightarrow x^{-3}y' = 4y^2 \Rightarrow y' = \frac{4y^2}{x^{-3}} \Rightarrow y' = 4x^3y^2$

Concept of Solution

A function

$h(=y)$ 與 $h'(=y')$ 代入 (4) 式中
滿足, 即 h 為 (4) 式的解

$y = h(x)$

is called a **solution** of a given ODE (4) on some open interval $a < x < b$ if $h(x)$ is defined and differentiable throughout the interval and is such that the equation becomes an identity if y and y' are replaced by h and h' , respectively. The curve (the graph) of h is called a **solution curve**.

Here, **open interval** $a < x < b$ means that the endpoints a and b are not regarded as points belonging to the interval. Also, $a < x < b$ includes *infinite intervals* $-\infty < x < b$, $a < x < \infty$, $-\infty < x < \infty$ (the real line) as special cases.

$y = h(x) + c$

如何求得呢? 以後章節會教...

EXAMPLE 1

Verification of Solution

Verify that $y = c/x$ (c an arbitrary constant) is a **solution** of the ODE $xy' = -y$ for all $x \neq 0$. Indeed, differentiate $y = c/x$ to get $y' = -c/x^2$. Multiply this by x , obtaining $xy' = -c/x$; thus, $xy' = -y$, the given ODE. ■

$y = \frac{c}{x} \Rightarrow y' = -\frac{c}{x^2} \Rightarrow xy' = -\frac{c}{x} = -y \Rightarrow xy' = -y$

EXAMPLE 2 Solution by Calculus. Solution Curves

The ODE $y' = dy/dx = \cos x$ can be solved directly by integration on both sides. Indeed, using calculus, we obtain $y = \int \cos x \, dx = \sin x + c$, where c is an arbitrary constant. This is a family of solutions. Each value of c , for instance, 2.75 or 0 or $-\pi$, gives one of these curves. Figure 3 shows some of them, for $c = -3, -2, -1, 0, 1, 2, 3, 4$.

$y = \sin x + c$ 為解，若是：

● c 為任意數： y 為通解(或一般解)(general solution)，很多條代表解的線。

● c 為固定常數： y 為特解(或特定解)(particular solution)，為一條特定的線。

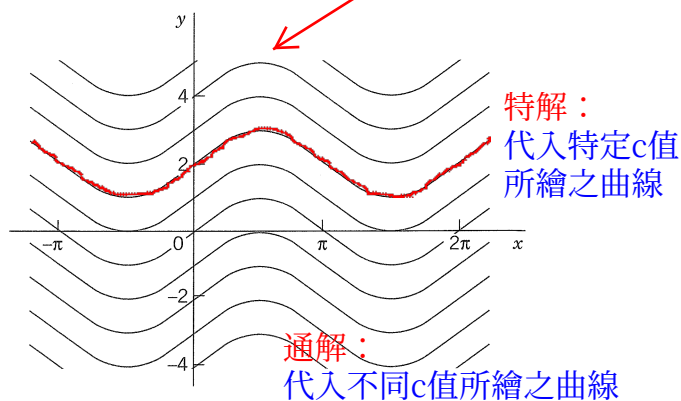


Fig. 3. Solutions $y = \sin x + c$ of the ODE $y' = \cos x$

EXAMPLE 3 (A) Exponential Growth. (B) Exponential Decay

From calculus we know that $y = ce^{0.2t}$ has the derivative

$$y' = \frac{dy}{dt} = 0.2e^{0.2t} = 0.2y.$$

Hence y is a solution of $y' = 0.2y$ (Fig. 4A). This ODE is of the form $y' = ky$. With positive-constant k it can model exponential growth, for instance, of colonies of bacteria or populations of animals. It also applies to humans for small populations in a large country (e.g., the United States in early times) and is then known as **Malthus's law**.¹ We shall say more about this topic in Sec. 1.5.

(B) Similarly, $y' = -0.2y$ (with a minus on the right) has the solution $y = ce^{-0.2t}$, (Fig. 4B) modeling exponential decay, as, for instance, of a radioactive substance (see Example 5).

$y' = ay$
 解 $y = ce^{at}$ (以後教)

$a > 0$, Exponential growth
 : 例如，細菌繁殖

$a < 0$, Exponential decay
 : 例如，放射性物質衰減

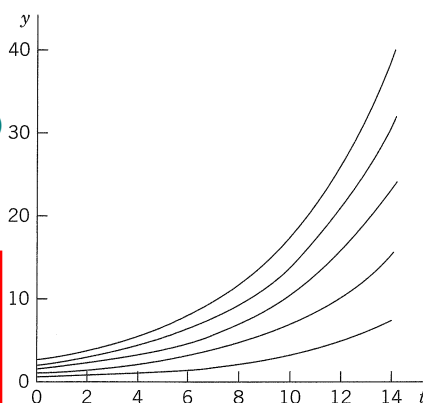


Fig. 4A. Solutions of $y' = 0.2y$ in Example 3 (exponential growth)

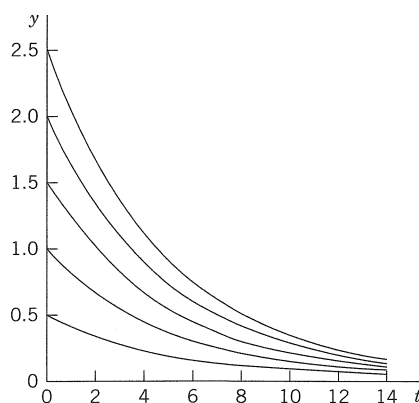


Fig. 4B. Solutions of $y' = -0.2y$ in Example 3 (exponential decay)

¹Named after the English pioneer in classic economics, THOMAS ROBERT MALTHUS (1766–1834).

$$y = \sin x + c \text{ (任意常數)}$$

We see that each ODE in these examples has a solution that contains an arbitrary constant c . Such a solution containing an arbitrary constant c is called a **general solution** of the ODE.

通解、一般解

(We shall see that c is sometimes not completely arbitrary but must be restricted to some interval to avoid complex expressions in the solution.)

We shall develop methods that will give general solutions *uniquely* (perhaps except for notation). Hence we shall say *the* general solution of a given ODE (instead of *a* general solution).

Geometrically, the general solution of an ODE is a family of infinitely many solution curves, one for each value of the constant c . If we choose a specific c (e.g., $c = 6.45$ or 0 or -2.01) we obtain what is called a **particular solution** of the ODE. A particular solution does not contain any arbitrary constants.

特解、特殊解、特定解

In most cases, general solutions exist, and every solution not containing an arbitrary constant is obtained as a particular solution by assigning a suitable value to c . Exceptions to these rules occur but are of minor interest in applications; see Prob. 16 in Problem Set 1.1.

初始值問題

$$y = \sin x + 3 \text{ (固定常數)}$$

Initial Value Problem

In most cases the unique solution of a given problem, hence a particular solution, is obtained from a general solution by an **initial condition** $y(x_0) = y_0$, with given values x_0 and y_0 , that is used to determine a value of the arbitrary constant c . Geometrically this condition means that the solution curve should pass through the point (x_0, y_0) in the xy -plane. An ODE, together with an initial condition, is called an **initial value problem**. Thus, if the ODE is explicit, $y' = f(x, y)$, the initial value problem is of the form

(5)

$$y' = f(x, y), \quad y(x_0) = y_0.$$

代入通解中，求得特定的 c 值，造成特解。一階ODE需要一組初始條件。

EXAMPLE 4 Initial Value Problem

Solve the initial value problem

$$y' = ay$$

$$\text{解 } y = ce^{at}$$

$$y' = \frac{dy}{dx} = 3y, \quad y(0) = 5.7.$$

通解、一般解

Solution. The general solution is $y(x) = ce^{3x}$; see Example 3. From this solution and the initial condition we obtain $y(0) = ce^0 = c = 5.7$. Hence the initial value problem has the solution $y(x) = 5.7e^{3x}$. This is a particular solution.

建模

特解、特定解

More on Modeling

The general importance of modeling to the engineer and physicist was emphasized at the beginning of this section. We shall now consider a basic physical problem that will show the details of the typical steps of modeling. **Step 1: the transition from the physical situation (the physical system) to its mathematical formulation (its mathematical model); Step 2: the solution by a mathematical method; and Step 3: the physical interpretation of the result.** This may be the easiest way to obtain a first idea of the nature and purpose of differential equations and their applications. Realize at the outset that your **computer** (your **CAS**) may perhaps give you a hand in Step 2, but Steps 1 and 3 are basically your work.

電腦無法幫忙，這是您的工作與必須建立的能力！

And Step 2 requires a solid knowledge and good understanding of solution methods available to you—you have to choose the method for your work by hand or by the computer. Keep this in mind, and always check computer results for errors (which may arise, for instance, from false inputs).

放射活動 指數衰減

EXAMPLE 5 Radioactivity. Exponential Decay

Given an amount of a radioactive substance, say, 0.5 g (gram), find the amount present at any later time.

Physical Information. Experiments show that at each instant a radioactive substance decomposes—and is thus decaying in time—proportional to the amount of substance present.

Step 1. Setting up a mathematical model of the physical process. Denote by $y(t)$ the amount of substance still present at any time t . By the physical law, the time rate of change $y'(t) = dy/dt$ is proportional to $y(t)$. This gives the *first-order ODE*

$$(6) \quad \frac{dy}{dt} = -ky \quad \leftarrow \text{建模}$$

where the constant k is positive, so that, because of the minus, we do get decay (as in [B] of Example 3). The value of k is known from experiments for various radioactive substances (e.g., $k = 1.4 \cdot 10^{-11} \text{ sec}^{-1}$, approximately, for radium $^{226}_{88}\text{Ra}$).

Now the given initial amount is 0.5 g, and we can call the corresponding instant $t = 0$. Then we have the *initial condition* $y(0) = 0.5$. This is the instant at which our observation of the process begins. It motivates the term *initial condition* (which, however, is also used when the independent variable is not time or when we choose a t other than $t = 0$). Hence the mathematical model of the physical process is the *initial value problem*

$$(7) \quad \frac{dy}{dt} = -ky, \quad y(0) = 0.5. \quad \leftarrow \text{初始值問題}$$

Step 2. Mathematical solution. As in (B) of Example 3 we conclude that the ODE (6) models exponential decay and has the general solution (with arbitrary constant c but definite given k)

$$(8) \quad y(t) = ce^{-kt}. \quad \leftarrow \text{求出通解 (以後教)}$$

We now determine c by using the initial condition. Since $y(0) = c$ from (8), this gives $y(0) = c = 0.5$. Hence the particular solution governing our process is (cf. Fig. 5)

$$(9) \quad \text{代入c值得特解} \rightarrow y(t) = 0.5e^{-kt} \quad \leftarrow \text{利用初始條件求出c值} > 0.$$

Always check your result—it may involve human or computer errors! Verify by differentiation (chain rule!) that your solution (9) satisfies (7) as well as $y(0) = 0.5$:

$$\frac{dy}{dt} = -0.5ke^{-kt} = -k \cdot 0.5e^{-kt} = -ky, \quad y(0) = 0.5e^0 = 0.5.$$

結果解釋

Step 3. Interpretation of result. Formula (9) gives the amount of radioactive substance at time t . It starts from the correct initial amount and decreases with time because k is positive. The limit of y as $t \rightarrow \infty$ is zero. ■

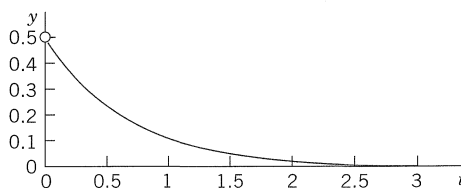


Fig. 5. Radioactivity (Exponential decay, $y = 0.5e^{-kt}$, with $k = 1.5$ as an example)

代表斜率

1.2 Geometric Meaning of $y' = f(x, y)$.

Direction Fields, Euler's Method

方向場：斜率方向的集合

A first-order ODE

點 (x, y) 斜率 $y'(x)$

(1)

 $y(x)$ 為解

$$y' = f(x, y)$$

值為 $f(x, y)$

has a simple geometric interpretation. From calculus you know that the derivative $y'(x)$ of $y(x)$ is the slope of $y(x)$. Hence a solution curve of (1) that passes through a point (x_0, y_0) must have, at that point, the slope $y'(x_0)$ equal to the value of f at that point; that is,

y 在點 x_0 處的斜率 $y'(x_0) = f(x_0, y_0)$. f 在點 (x_0, y_0) 的值

Using this fact, we can develop graphic or numeric methods for obtaining approximate solutions of ODEs (1). This will lead to a better conceptual understanding of an ODE (1). Moreover, such methods are of practical importance since many ODEs have complicated solution formulas or no solution formulas at all, whereby numeric methods are needed.

用手繪法或電腦繪圖獲得解 (y) 之曲線

Graphic Method of Direction Fields. Practical Example Illustrated in Fig. 7. We can show directions of solution curves of a given ODE (1) by drawing short straight-line segments (lineal elements) in the xy -plane. This gives a direction field (or slope field) into which lineal elements fit (approximate) solution curves. This may reveal typical properties of the whole family of solutions.

Figure 7 shows a direction field for the ODE

(2)

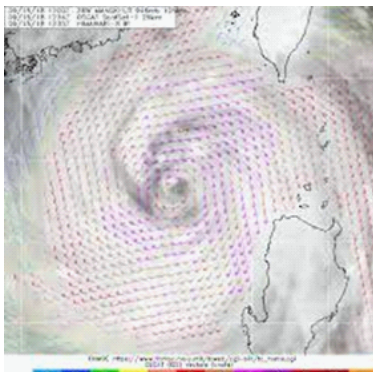
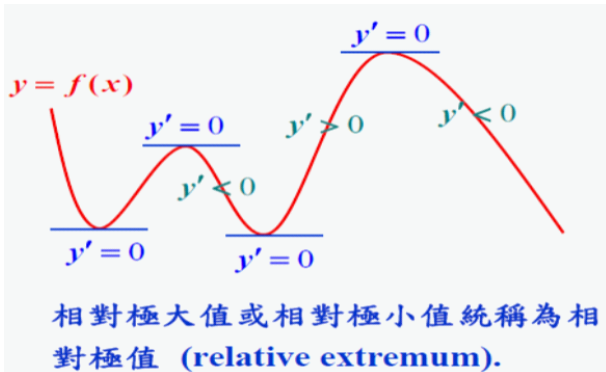
$$y' = y + x$$

(Computer Algebra System) and some approximate solution curves

 x, y 值的變化造成線性元素斜率的改變

線性元素的集合

由所知的切線方向 $y'(x)$ 與通過點 (x, y) ，繪出趨近解之曲線。



某颱風之風場圖

Fig. 7. Direction field of $y' = y + x$, with three approximate solution curves passing through $(0, 1)$, $(0, 0)$, $(0, -1)$, respectively

可用手繪法或電腦繪圖獲得解 (y) 之曲線，但利用工數法求解才是聰明的方法。因為有的方程式很複雜，無法輕易以手繪法獲得 $y(x)$ 解曲線。

If you have no CAS, first draw a few *level curves* $f(x, y) = \text{const}$ of $f(x, y)$, then parallel lineal elements along each such curve (which is also called an *isocline*, meaning a curve of equal inclination), and finally draw approximation curves fit to the lineal elements.

We shall now illustrate how numeric methods work by applying the simplest numeric method, that is Euler's method, to an initial value problem involving ODE (2). First we give a brief description of Euler's method.

Numeric Method by Euler

Given an ODE (1) and an initial value $y(x_0) = y_0$, **Euler's method** yields approximate solution values at equidistant x -values $x_0, x_1 = x_0 + h, x_2 = x_0 + 2h, \dots$, namely,

$$y_1 = y_0 + hf(x_0, y_0) \quad (\text{Fig. 8})$$

$$y_2 = y_1 + hf(x_1, y_1), \quad \text{etc.}$$

In general,

$$y_n = y_{n-1} + hf(x_{n-1}, y_{n-1})$$

where the step h equals, e.g., 0.1 or 0.2 (as in Table 1.1) or a smaller value for greater accuracy.

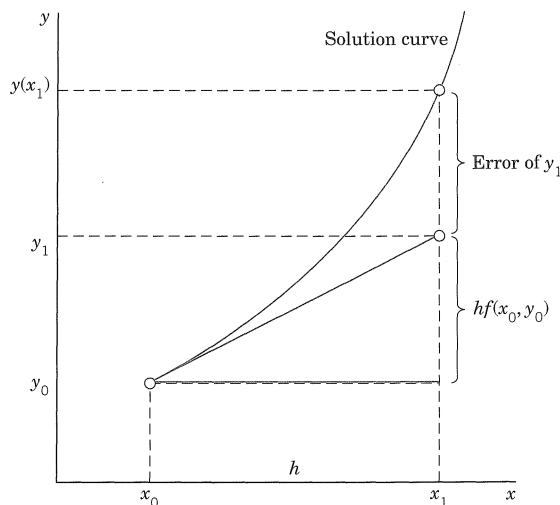


Fig. 8. First Euler step, showing a solution curve, its tangent at (x_0, y_0) , step h and increment $hf(x_0, y_0)$ in the formula for y_1

Table 1.1 shows the computation of $n = 5$ steps with step $h = 0.2$ for the ODE (2) and initial condition $y(0) = 0$, corresponding to the middle curve in the direction field. We shall solve the ODE exactly in Sec. 1.5. For the time being, verify that the initial value problem has the solution $y = e^x - x - 1$. The solution curve and the values in Table 1.1 are shown in Fig. 9. These values are rather inaccurate. The errors $y(x_n) - y_n$ are shown in Table 1.1 as well as in Fig. 9. Decreasing h would improve the values, but would soon require an impractical amount of computation. Much better methods of a similar nature will be discussed in Sec. 21.1.

題目 $y' = ay$

[解]

$$\frac{dy}{dt} = ay \dots \dots \dots \text{步驟1: } y' \text{ 以 } \frac{dy}{dt} \text{ 替代}$$

$$\frac{dy}{y} = a dt \dots \dots \dots \text{步驟2: } y \text{ 與 } t \text{ 要分開於等號兩邊}$$

$$\int \frac{dy}{y} = \int a dt + c^* \dots \dots \dots \text{步驟3: 等號兩邊取積分，再加 } c^*$$

$$\ln y = at + c^* \Rightarrow y = e^{at + c^*} = e^{at} e^{c^*} = c e^{at} \dots \dots \text{步驟4: 積分技巧的熟練}$$

$$\text{答案 } y = c e^{at}$$

cally solving
blem (IVP).
principle are
get a feel for
olve the IVP
s. Graph the
on the same

1.3 (變數)可分離的 Separable ODEs. Modeling

學習重點：

1. 分離變數解題法
2. 學會如何建模

Many practically useful ODEs can be reduced to the form

$$g(y)y' = f(x) \quad \text{① } y' = \frac{dy}{dx}$$

(1)

by purely algebraic manipulations. Then we can integrate on both sides with respect to x , obtaining

$$(2) \quad \int g(y)y' dx = \int f(x) dx + c.$$

等號兩邊對 x 積分

On the left we can switch to y as the variable of integration. By calculus, $y' dx = dy$, so that

$$(3) \quad \int g(y) dy = \int f(x) dx + c.$$

此積分值即為 eq(1) 的一般解

等號兩邊取積分(再加 c)

If f and g are continuous functions, the integrals in (3) exist, and by evaluating them we obtain a general solution of (1). This method of solving ODEs is called the **method of separating variables**, and (1) is called a **separable equation**, because in (3) the variables are now separated: x appears only on the right and y only on the left.

分離變數法

EXAMPLE 1

Separable ODE

The ODE $y' = 1 + y^2$ is separable because it can be written

$$\frac{dy}{1+y^2} = dx + c$$

By integration,

$$\arctan y = x + c$$

or

$$y = \tan(x + c)$$

特別小心！

It is very important to introduce the constant of integration immediately when the integration is performed. If we wrote $\arctan y = x$, then $y = \tan x$, and then introduced c , we would have obtained $y = \tan x + c$, which is not a solution (when $c \neq 0$). Verify this.

注意不同點！

$$\int x e^x + e^x dx = x e^x - e^{-x} - e^{-x} = -(\pi+2)e^{-x} \quad 13$$

↓ 分部積分 ↑

分部積分為一重要技巧，務必熟練！

EXAMPLE 2

Separable ODE $\frac{dy}{dx}$

The ODE $y' = (x+1)e^{-x}y^2$ is separable; we obtain $y^{-2} dy = (x+1)e^{-x} dx$. $+C$

By integration, $-y^{-1} = \frac{-(x+2)e^{-x}}{1} + c$, $y = \frac{1}{(x+2)e^{-x} - c}$.

$$\int u dv = uv - \int v du$$

EXAMPLE 3

Initial Value Problem (IVP). Bell-Shaped Curve

Solve $y' = -2xy$, $y(0) = 1.8$.

Solution. By separation and integration,

$$y = e^{-x^2 + \tilde{c}} = e^{-x^2} \cdot e^{\tilde{c}} = c e^{-x^2}$$

通解

This is the general solution. From it and the initial condition, $y(0) = ce^0 = c = 1.8$. Hence the IVP has the solution $y = 1.8e^{-x^2}$. This is a particular solution, representing a bell-shaped curve (Fig. 10).

特解

$$\begin{aligned} \int x e^x dx &= -x e^{-x} + \int e^{-x} dx \\ &= -x e^{-x} - e^{-x} \end{aligned}$$

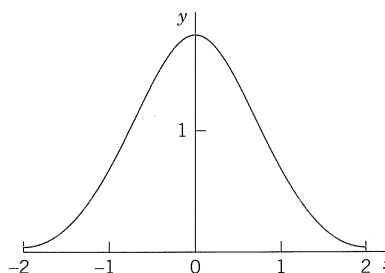


Fig. 10. Solution in Example 3 (bell-shaped curve)

Modeling

The importance of modeling was emphasized in Sec. 1.1, and separable equations yield various useful models. Let us discuss this in terms of some typical examples.

EXAMPLE 4

Radiocarbon Dating² 放射性碳定年法

In September 1991 the famous Iceman (Oetzi), a mummy from the Neolithic period of the Stone Age found in the ice of the Oetztal Alps (hence the name “Oetzi”) in Southern Tyrolia near the Austrian–Italian border, caused a scientific sensation. When did Oetzi approximately live and die if the ratio of carbon $^{14}_6\text{C}$ to carbon $^{12}_6\text{C}$ in this mummy is 52.5% of that of a living organism?

Physical Information. In the atmosphere and in living organisms, the ratio of radioactive carbon $^{14}_6\text{C}$ (made radioactive by cosmic rays) to ordinary carbon $^{12}_6\text{C}$ is constant. When an organism dies, its absorption of $^{14}_6\text{C}$ by breathing and eating terminates. Hence one can estimate the age of a fossil by comparing the radioactive carbon ratio in the fossil with that in the atmosphere. To do this, one needs to know the half-life of $^{14}_6\text{C}$, which is 5715 years (*CRC Handbook of Chemistry and Physics*, 83rd ed., Boca Raton: CRC Press, 2002, page 11–52, line 9).

Solution. **Modeling.** Radioactive decay is governed by the ODE $y' = ky$ (see Sec. 1.1, Example 5). By separation and integration (where t is time and y_0 is the initial ratio of $^{14}_6\text{C}$ to $^{12}_6\text{C}$)

$$\frac{dy}{y} = k dt, \quad \ln |y| = kt + c, \quad y = y_0 e^{kt} \quad (y_0 = e^c).$$

²Method by WILLARD FRANK LIBBY (1908–1980), American chemist, who was awarded for this work the 1960 Nobel Prize in chemistry.

Next we use the half-life $H = 5715$ to determine k . When $t = H$, half of the original substance is still present. Thus,

$$y_0 e^{kH} = 0.5y_0, \quad e^{kH} = 0.5, \quad k = \frac{\ln 0.5}{H} = \frac{-0.693}{5715} = -0.0001213.$$

Finally, we use the ratio 52.5% for determining the time t when Oetzi died (actually, was killed),

$$e^{kt} = e^{-0.0001213t} = 0.525, \quad t = \frac{\ln 0.525}{-0.0001213} = 5312. \quad \text{Answer:} \quad \text{About 5300 years ago.}$$

Other methods show that radiocarbon dating values are usually too small. According to recent research, this is due to a variation in that carbon ratio because of industrial pollution and other factors, such as nuclear testing. ■

EXAMPLE 5

Mixing Problem

Mixing problems occur quite frequently in chemical industry. We explain here how to solve the basic model involving a single tank. The tank in Fig. 11 contains 1000 gal of water in which initially 100 lb of salt is dissolved. Brine runs in at a rate of 10 gal/min, and each gallon contains 5 lb of dissolved salt. The mixture in the tank is kept uniform by stirring. Brine runs out at 10 gal/min. Find the amount of salt in the tank at any time t .

Solution. *Step 1. Setting up a model.* Let $y(t)$ denote the amount of salt in the tank at time t . Its time rate of change is

$$y' = \text{Salt inflow rate} - \text{Salt outflow rate} \quad \text{Balance law.}$$

5 lb times 10 gal gives an inflow of 50 lb of salt. Now, the outflow is 10 gal of brine. This is $10/1000 = 0.01$ ($= 1\%$) of the total brine content in the tank, hence 0.01 of the salt content $y(t)$, that is, $0.01 y(t)$. Thus the model is the ODE

$$(4) \quad y' = 50 - 0.01y = -0.01(y - 5000).$$

Step 2. Solution of the model. The ODE (4) is separable. Separation, integration, and taking exponents on both sides gives

$$\frac{dy}{y - 5000} = -0.01 dt, \quad \ln |y - 5000| = -0.01t + c^*, \quad y - 5000 = ce^{-0.01t}.$$

Initially the tank contains 100 lb of salt. Hence $y(0) = 100$ is the initial condition that will give the unique solution. Substituting $y = 100$ and $t = 0$ in the last equation gives $100 - 5000 = ce^0 = c$. Hence $c = -4900$. Hence the amount of salt in the tank at time t is

$$(5) \quad y(t) = 5000 - 4900e^{-0.01t}.$$

This function shows an exponential approach to the limit 5000 lb; see Fig. 11. Can you explain physically that $y(t)$ should increase with time? That its limit is 5000 lb? Can you see the limit directly from the ODE?

The model discussed becomes more realistic in problems on pollutants in lakes (see Problem Set 1.5, Prob. 35) or drugs in organs. These types of problems are more difficult because the mixing may be imperfect and the flow rates (in and out) may be different and known only very roughly. ■

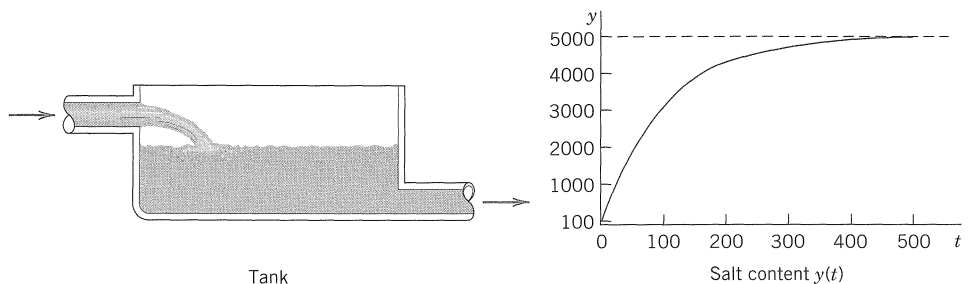


Fig. 11. Mixing problem in Example 5

EXAMPLE 6 Heating an Office Building (Newton's Law of Cooling)³**室內溫度變化：****P.M. 10 → T(0)=70 °F (約21 °C)****A.M. 02 → T(4)=65 °F**

(初始條件)

A.M. 06 → T(8)=?? °F**室外溫度變化：****P.M. 10 → 50 °F (約10 °C)****A.M. 06 → 40 °F**

Suppose that in winter the daytime temperature in a certain office building is maintained at 70°F. The heating is shut off at 10 P.M. and turned on again at 6 A.M. On a certain day the temperature inside the building at 2 A.M. was found to be 65°F. The outside temperature was 50°F at 10 P.M. and had dropped to 40°F by 6 A.M. What was the temperature inside the building when the heat was turned on at 6 A.M.?

Physical information Experiments show that the time rate of change of the temperature T of a body B (which conducts heat well, for example, as a copper ball does) is proportional to the difference between T and the temperature of the surrounding medium (Newton's law of cooling).

Solution. Step 1. Setting up a model. Let $T(t)$ be the temperature inside the building and T_A the outside temperature (assumed to be constant in Newton's law). Then by Newton's law,

$$(6) \quad T' = \frac{dT}{dt} = k(T - T_A).$$

物體溫度 (室內溫度)
環境溫度 (取45 °F)

Such experimental laws are derived under idealized assumptions that rarely hold exactly. However, even if a model seems to fit the reality only poorly (as in the present case), it may still give valuable qualitative information. To see how good a model is, the engineer will collect experimental data and compare them with calculations from the model.

Step 2. General solution. We cannot solve (6) because we do not know T_A , just that it varied between 50°F and 40°F, so we follow the **Golden Rule**: If you cannot solve your problem, try to solve a simpler one. We solve (6) with the unknown function T_A replaced with the average of the two known values, or 45°F. For physical reasons we may expect that this will give us a reasonable approximate value of T in the building at 6 A.M.

For constant $T_A = 45$ (or any other constant value) the ODE (6) is separable. Separation, integration, and taking exponents gives the general solution

$$\int \frac{dT}{T - 45} = \int k dt, \quad \ln |T - 45| = kt + c^*, \quad T(t) = 45 + ce^{kt} \quad (c = e^{c^*}).$$

通解

Step 3. Particular solution. We choose 10 P.M. to be $t = 0$. Then the given initial condition is $T(0) = 70$ and yields a particular solution, call it T_p . By substitution,

$$T(0) = 45 + ce^0 = 70, \quad c = 70 - 45 = 25, \quad T_p(t) = 45 + 25e^{kt}.$$

特解 求c值

Step 4. Determination of k . We use $T(4) = 65$, where $t = 4$ is 2 A.M. Solving algebraically for k and inserting k into $T_p(t)$ gives (Fig. 12)

$$T_p(4) = 45 + 25e^{4k} = 65, \quad e^{4k} = 0.8, \quad k = \frac{1}{4} \ln 0.8 = -0.056, \quad T_p(t) = 45 + 25e^{-0.056t}.$$

求k值

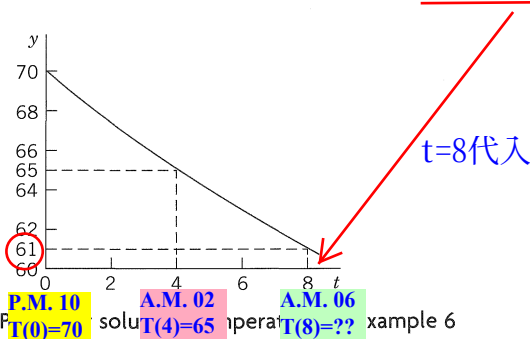


Fig. 12. P.M. 10 T(0)=70 A.M. 02 T(4)=65 A.M. 06 T(8)=?? Example 6

³Sir ISAAC NEWTON (1642–1727), great English physicist and mathematician, became a professor at Cambridge in 1669 and Master of the Mint in 1699. He and the German mathematician and philosopher GOTTFRIED WILHELM LEIBNIZ (1646–1716) invented (independently) the differential and integral calculus. Newton discovered many basic physical laws and created the method of investigating physical problems by means of calculus. His *Philosophiæ naturalis principia mathematica* (*Mathematical Principles of Natural Philosophy*, 1687) contains the development of classical mechanics. His work is of greatest importance to both mathematics and physics.

Step 5. Answer and interpretation. 6 A.M. is $t = 8$ (namely, 8 hours after 10 P.M.), and

$$T_1(8) = 45 + 25e^{-0.056 \cdot 8} = 61[^\circ\text{F}].$$

Hence the temperature in the building dropped 9°F , a result that looks reasonable. ■

EXAMPLE 7

Leaking Tank. Outflow of Water Through a Hole (Torricelli's Law)

This is another prototype engineering problem that leads to an ODE. It concerns the outflow of water from a cylindrical tank with a hole at the bottom (Fig. 13). You are asked to find the height of the water in the tank at any time if the tank has diameter 2 m, the hole has diameter 1 cm, and the initial height of the water when the hole is opened is 2.25 m. When will the tank be empty?

Physical information. Under the influence of gravity the outflowing water has velocity

$$(7) \quad v(t) = 0.600\sqrt{2gh(t)} \quad (\text{Torricelli's law}^4),$$

where $h(t)$ is the height of the water above the hole at time t , and $g = 980 \text{ cm/sec}^2 = 32.17 \text{ ft/sec}^2$ is the acceleration of gravity at the surface of the earth.

Solution. *Step 1. Setting up the model.* To get an equation, we relate the decrease in water level $h(t)$ to the outflow. The volume ΔV of the outflow during a short time Δt is

$$\Delta V = Av \Delta t \quad (A = \text{Area of hole}).$$

ΔV must equal the change ΔV^* of the volume of the water in the tank. Now

$$\Delta V^* = -B \Delta h \quad (B = \text{Cross-sectional area of tank})$$

where $\Delta h (> 0)$ is the decrease of the height $h(t)$ of the water. The minus sign appears because the volume of the water in the tank decreases. Equating ΔV and ΔV^* gives

$$-B \Delta h = Av \Delta t.$$

We now express v according to Torricelli's law and then let Δt (the length of the time interval considered) approach 0—this is a *standard way* of obtaining an ODE as a model. That is, we have

$$\frac{\Delta h}{\Delta t} = -\frac{A}{B}v = -\frac{A}{B}0.600\sqrt{2gh(t)}$$

and by letting $\Delta t \rightarrow 0$ we obtain the ODE

$$\frac{dh}{dt} = -26.56 \frac{A}{B} \sqrt{h},$$

where $26.56 = 0.600\sqrt{2 \cdot 980}$. This is our model, a first-order ODE.

Step 2. General solution. Our ODE is separable. A/B is constant. Separation and integration gives

$$\frac{dh}{\sqrt{h}} = -26.56 \frac{A}{B} dt \quad \text{and} \quad 2\sqrt{h} = c^* - 26.56 \frac{A}{B} t.$$

Dividing by 2 and squaring gives $h = (c - 13.28At/B)^2$. Inserting $13.28A/B = 13.28 \cdot 0.5^2\pi/100^2\pi = 0.000332$ yields the general solution

$$h(t) = (c - 0.000332t)^2.$$

⁴EVANGELISTA TORRICELLI (1608–1647), Italian physicist, pupil and successor of GALILEO GALILEI (1564–1642) at Florence. The “contraction factor” 0.600 was introduced by J. C. BORDA in 1766 because the stream has a smaller cross section than the area of the hole.

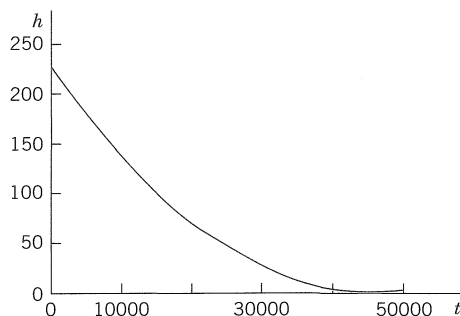
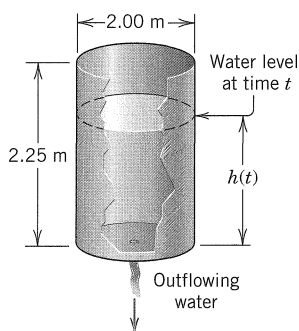
Step 3. Particular solution. The initial height (the initial condition) is $h(0) = 225$ cm. Substitution of $t = 0$ and $h = 225$ gives from the general solution $c^2 = 225$, $c = 15.00$ and thus the particular solution (Fig. 13)

$$h_p(t) = (15.00 - 0.000332t)^2.$$

Step 4. Tank empty. $h_p(t) = 0$ if $t = 15.00/0.000332 = 45,181$ [sec] = 12.6 [hours].

Here you see distinctly the **importance of the choice of units**—we have been working with the cgs system, in which time is measured in seconds! We used $g = 980$ cm/sec².

Step 5. Checking. Check the result.



Water level $h(t)$ in tank

flow from a cylindrical tank ("leaking tank").
Torricelli's law

學習重點：

遇見非分離變數之標準題型時，該如何進行化簡轉換？

以 $u=y/x$ 進行替換的題型！

Extended Method: Reduction to Separable Form

Certain nonseparable ODEs can be made separable by transformations that introduce for y a new unknown function. We discuss this technique for a class of ODEs of practical importance, namely, for equations

$\frac{y}{x}$ as

step4: 整理成可分離變數型式
(8)

$$y' = f\left(\frac{y}{x}\right).$$

step1: 必須先化成有 y/x 的形式，再令 $u=y/x$ 代入

$$u'x + u = f(u)$$

$$\frac{du}{dx}x = f(u) - u$$

$$\frac{du}{f(u)-u} = \frac{dx}{x}$$

Here, f is any (differentiable) function of y/x , such as $\sin(y/x)$, $(y/x)^4$, and so on. (Such an ODE is sometimes called a *homogeneous ODE*, a term we shall not use but reserve for a more important purpose in Sec. 1.5.)

The form of such an ODE suggests that we set $y/x = u$; thus,

step2: 令

$$(9) \quad y = ux$$

and by product differentiation

取微分

$$y' = u'x + u.$$

Substitution into $y' = f(y/x)$ then gives $u'x + u = f(u)$ or $u'x = f(u) - u$. We see that if $f(u) - u \neq 0$, this can be separated:

(10) 變成可分離變數題型

$$\frac{du}{f(u) - u} = \frac{dx}{x}$$

$$\rightarrow \int \frac{du}{f(u)-u} = \int \frac{dx}{x} + C$$

step5: 解可分離變數型式，求出 u

可求得 u ，但 $u=y/x$ ，故 $y=ux$ 才是正確答案

EXAMPLE 8 Reduction to Separable Form

Solve

$$2xyy' = y^2 - x^2.$$

Solution. To get the usual explicit form, divide the given equation by $2xy$,

$$y' = \frac{y^2 - x^2}{2xy} = \frac{y}{2x} - \frac{x}{2y}$$

Now substitute y and y' from (9) and then simplify by subtracting u on both sides,

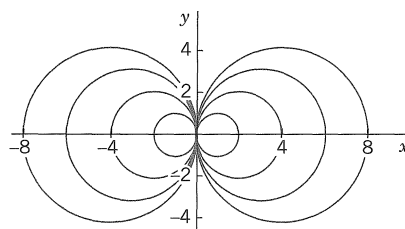
$$\frac{u'x + u}{u} = \frac{u}{2} - \frac{1}{2u}, \quad \frac{u'}{u} = -\frac{u}{2} - \frac{1}{2u} = \frac{-u^2 - 1}{2u}.$$

You see that in the last equation you can now separate the variables,

$$\int \frac{2u \, du}{1 + u^2} = \int \frac{-dx}{x} + C^* \quad \text{By integration,} \quad \ln(1 + u^2) = -\ln|x| + C^* = \ln\left|\frac{1}{x}\right| + C^*.$$

Take exponents on both sides to get $1 + u^2 = c/x$ or $1 + (y/x)^2 = c/x$. Multiply the last equation by x^2 to obtain (Fig. 14)

$$x^2 + y^2 = cx. \quad \text{Thus} \quad \left(x - \frac{c}{2}\right)^2 + y^2 = \frac{c^2}{4} \quad \text{圓方程式}$$

This general solution represents a family of circles passing through the origin with centers on the x -axis. ■**Fig. 14.** General solution (family of circles) in Example 8

代入不同c值所繪之圖

PROBLEM SET 1.3**1. CAUTION! Constant of integration.** Why is it important to introduce the constant of integration immediately when you integrate?**2-10** GENERAL SOLUTION

Find a general solution. Show the steps of derivation. Check your answer by substitution.

2. $y^3 y' + x^3 = 0$
3. $y' = \sec^2 y$
4. $y' \sin 2\pi x = \pi y \cos 2\pi x$
5. $yy' + 36x = 0$
6. $y' = e^{2x-1} y^2$
7. $xy' = y + 2x^3 \sin^2 \frac{y}{x}$ (Set $y/x = u$)
8. $y' = (y + 4x)^2$ (Set $y + 4x = v$)
9. $xy' = y^2 + y$ (Set $y/x = u$)
10. $xy' = x + y$ (Set $y/x = u$)

多做習題
增加解題
熟練度!**11-17** INITIAL VALUE PROBLEMS (IVPs)

Solve the IVP. Show the steps of derivation, beginning with the general solution.

11. $xy' + y = 0, \quad y(4) = 6$
12. $y' = 1 + 4y^2, \quad y(1) = 0$
13. $y' \cosh^2 x = \sin^2 y, \quad y(0) = \frac{1}{2}\pi$
14. $dr/dt = -2tr, \quad r(0) = r_0$
15. $y' = -4x/y, \quad y(2) = 3$
16. $y' = (x + y - 2)^2, \quad y(0) = 2$
(Set $v = x + y - 2$)
17. $xy' = y + 3x^4 \cos^2(y/x), \quad y(1) = 0$
(Set $y/x = u$)

18. Particular solution. Introduce limits of integration in (3) such that y obtained from (3) satisfies the initial condition $y(x_0) = y_0$.

- 33. Rope.** To tie a boat in a harbor, how many times must a rope be wound around a bollard (a vertical rough cylindrical post fixed on the ground) so that a man holding one end of the rope can resist a force exerted by the boat 1000 times greater than the man can exert? First guess. Experiments show that the change ΔS of the force S in a small portion of the rope is proportional to S and to the small angle $\Delta\phi$ in Fig. 16. Take the proportionality constant 0.15. The result should surprise you!

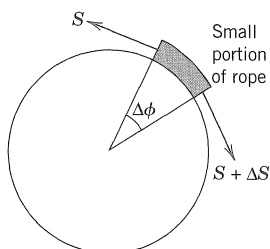


Fig. 16. Problem 33

- 34. TEAM PROJECT. Family of Curves.** A family of curves can often be characterized as the general solution of $y' = f(x, y)$.
- Show that for the circles with center at the origin we get $y' = -x/y$.
 - Graph some of the hyperbolas $xy = c$. Find an ODE for them.
 - Find an ODE for the straight lines through the origin.
 - You will see that the product of the right sides of the ODEs in (a) and (c) equals -1 . Do you recognize

this as the condition for the two families to be orthogonal (i.e., to intersect at right angles)? Do your graphs confirm this?

(e) Sketch families of curves of your own choice and find their ODEs. Can every family of curves be given by an ODE?

- 35. CAS PROJECT. Graphing Solutions.** A CAS can usually graph solutions, even if they are integrals that cannot be evaluated by the usual analytical methods of calculus.

(a) Show this for the five initial value problems $y' = e^{-x^2}$, $y(0) = 0, \pm 1, \pm 2$, graphing all five curves on the same axes.

(b) Graph approximate solution curves, using the first few terms of the Maclaurin series (obtained by term-wise integration of that of y') and compare with the exact curves.

(c) Repeat the work in (a) for another ODE and initial conditions of your own choice, leading to an integral that cannot be evaluated as indicated.

- 36. TEAM PROJECT. Torricelli's Law.** Suppose that the tank in Example 7 is hemispherical, of radius R , initially full of water, and has an outlet of 5 cm^2 cross-sectional area at the bottom. (Make a sketch.) Set up the model for outflow. Indicate what portion of your work in Example 7 you can use (so that it can

學習重點：

1. 何謂正合ODE? (必須先確定題型)

2. 如何求解正合ODE?

3. 碰見非正合ODE該怎麼辦?

非正合ODE $\times$ 積分因子 $\rightarrow$ 正合ODE

4. 如何求出積分因子?

1.4 Exact ODEs. Integrating Factors

推導公式

$$u(x(t), y(t)) = c$$

$$\frac{du}{dt} = \frac{\partial u}{\partial x} \frac{dx}{dt} + \frac{\partial u}{\partial y} \frac{dy}{dt} = 0$$

$$\Rightarrow du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy = 0$$

$$\text{or } \frac{dy}{dx} = - \frac{\frac{\partial u}{\partial x}}{\frac{\partial u}{\partial y}}$$

We recall from calculus that if a function $u(x, y)$ has continuous partial derivatives, its differential (also called its total differential) is

全微分

偏微分

$$du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy$$

解題前提....

From this it follows that if $u(x, y) = c = \text{const}$, then $du = 0$.

For example, if $u = x + x^2 y^3 = c$, then

解答

$$du = (1 + 2xy^3) dx + 3x^2 y^2 dy = 0$$

如何求出?

若ODE已知

$$y' = \frac{dy}{dx} = - \frac{1 + 2xy^3}{3x^2 y^2}$$

題目出現的形式

$$y' = - \frac{M(x, y)}{N(x, y)}$$

求出解 $u(x, y) = c$

解題程序

上述觀念引發以下正合ODE解法

an ODE that we can solve by going backward. This idea leads to a powerful solution method as follows.

A first-order ODE $M(x, y) + N(x, y)y' = 0$, written as (use $dy = y'dx$ as in Sec. 1.3)

(1) 注意標準型式 $M(x, y) dx + N(x, y) dy = 0$ 重要！！

is called an **exact differential equation** if the differential form $M(x, y) dx + N(x, y) dy$ is **exact**, that is, this form is the differential

$$(2) \quad du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy \stackrel{\triangle}{=} 0$$

of some function $u(x, y)$. Then (1) can be written

$$du = 0.$$

By integration we immediately obtain the general solution of (1) in the form

(3) $u(x, y) = c.$ 函數u等於一常數，此即為待求解

This is called an **implicit solution**, in contrast to a solution $y = h(x)$ as defined in Sec. 1.1, which is also called an *explicit solution*, for distinction. Sometimes an implicit solution can be converted to explicit form. (Do this for $x^2 + y^2 = 1$.) If this is not possible, your CAS may graph a figure of the **contour lines** (3) of the function $u(x, y)$ and help you in understanding the solution.

Comparing (1) and (2), we see that (1) is an exact differential equation if there is some function $u(x, y)$ such that

(4) 重要！！ $(a) \quad \frac{\partial u}{\partial x} = M, \quad (b) \quad \frac{\partial u}{\partial y} = N.$ 重要條件，可用以求u

From this we can derive a formula for checking whether (1) is exact or not, as follows.

Let M and N be continuous and have continuous first partial derivatives in a region in the xy -plane whose boundary is a closed curve without self-intersections. Then by partial differentiation of (4) (see App. 3.2 for notation),

$$\begin{aligned} \frac{\partial M}{\partial y} &= \frac{\partial^2 u}{\partial y \partial x} \\ \frac{\partial N}{\partial x} &= \frac{\partial^2 u}{\partial x \partial y} \end{aligned} \quad \left. \begin{array}{l} \text{兩式相等} \end{array} \right\}$$

By the assumption of continuity the two second partial derivatives are equal. Thus

(5) 重要！！ $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

此為(1)式滿足正合ODE的充分且必要條件，必須滿足此一條件，才代表(1)式為正合ODE形式，必有 $u(x, y)=c$ 的解。

題目型式：

$M(x, y)dx + N(x, y)dy = 0 \dots (1)$

確認(1)式為正合ODE
後，才可以用以下的方法求解，得到
 $u(x,y)=c$ 的解。

This condition is not only necessary but also sufficient for (1) to be an exact differential equation. (We shall prove this in Sec. 10.2 in another context. Some calculus books, for instance, [GenRef 12], also contain a proof.)

If (1) is exact, the function $u(x, y)$ can be found by inspection or in the following systematic way. From (4a) we have by integration with respect to x

$$(6) \quad \frac{\partial u}{\partial x} = M \rightarrow u = \int M dx + k(y); \quad \text{等號兩邊、比較項式得 } k(y)$$

$$\rightarrow \frac{\partial u}{\partial y} = \frac{\partial}{\partial y} \left[\int M dx \right] + \frac{dk(y)}{dy} = N$$

in this integration, y is to be regarded as a constant, and $k(y)$ plays the role of a “constant” of integration. To determine $k(y)$, we derive $\partial u / \partial y$ from (6), use (4b) to get dk/dy , and use (4b) to get k . (See Example 1, below.)

Formula (6) was obtained from (4a). Instead of (4a) we may equally well use (4b). Then, instead of (6), we first have by integration with respect to y

$$(6^*) \quad \frac{\partial u}{\partial y} = N \rightarrow u = \int N dy + l(x); \quad \text{等號兩邊、比較項式得 } l(x)$$

$$\rightarrow \frac{\partial u}{\partial x} = \frac{\partial}{\partial x} \left[\int N dy \right] + \frac{dl(x)}{dx} = M$$

To determine $l(x)$, we derive $\partial u / \partial x$ from (6*), use (4a) to get dl/dx , and integrate. We illustrate all this by the following typical examples.

題型： $M(x, y)dx + N(x, y)dy = 0$

正合ODE解題步驟：

1. 找出 $M(x, y)$ 與 $N(x, y)$

$$\frac{\partial u}{\partial x} = M$$

$$\frac{\partial u}{\partial y} = N$$

2. 判斷是否為正合型式

$$\frac{\partial M(x, y)}{\partial y} = \frac{\partial N(x, y)}{\partial x} \quad (\text{正合})$$

$$\frac{\partial M(x, y)}{\partial y} \neq \frac{\partial N(x, y)}{\partial x} \quad (\text{非正合}) \Rightarrow \text{求積分因子(下節教)}$$

$$3. \frac{\partial u}{\partial x} = M \Rightarrow u = \int M dx + k(y) \dots \dots (A \text{式})$$

$$4. \frac{\partial u}{\partial y} = N = \frac{\partial [A \text{式}]}{\partial y} = \frac{\partial [\int M dx + k(y)]}{\partial y} \Rightarrow \text{比較等號兩邊項式，求出 } k(y)$$

$$5. k(y) \text{帶回}(A \text{式}) \cdot u = \int M dx + k(y) = c \text{ 為答案}$$

或

$$6. \frac{\partial u}{\partial y} = N \Rightarrow u = \int N dy + l(x) \dots \dots (B \text{式})$$

$$7. \frac{\partial u}{\partial x} = M = \frac{\partial [B \text{式}]}{\partial x} = \frac{\partial [\int N dy + l(x)]}{\partial x} \Rightarrow \text{比較等號兩邊項式，求出 } l(x)$$

$$8. l(x) \text{帶回}(B \text{式}) \cdot u = \int N dy + l(x) = c \text{ 為答案}$$

If $u(x, y) = c \Rightarrow du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy = 0 = Mdx + Ndy$ (或由 $y' = \frac{dy}{dx} = \frac{-M}{N}$ 化簡)

ie,
$$\begin{cases} \frac{\partial u}{\partial x} = M \dots\dots (a) \\ \frac{\partial u}{\partial y} = N \dots\dots (b) \end{cases}$$

題目的型式

解題步驟：

(1) 確認是否為正合ODE： $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ ($\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$) (滿足正合ODE的充分且必要條件)

(2-1) From (a) 求u

$\Rightarrow u = \int M dx + k(y) = \text{積分1} + k(y) \dots\dots (c)$

(c) 對y微分 $\Rightarrow \frac{\partial u}{\partial y} = \frac{\partial}{\partial y} [\text{積分1}] + \frac{d}{dy} [k(y)] = N$ (比較等號兩邊項式，求出k(y)，再帶入(c)式)

ANS-1: 積分1 + k(y) = c (一定要記得加上)

(2-2) From (b) 求u

$\Rightarrow u = \int N dy + l(x) = \text{積分2} + l(x) \dots\dots (d)$

(d) 對x微分 $\Rightarrow \frac{\partial u}{\partial x} = \frac{\partial}{\partial x} [\text{積分2}] + \frac{d}{dx} [l(x)] = M$ (比較等號兩邊項式，求出l(x)，再帶入(d)式)

ANS-2: 積分2 + l(x) = c (答案一定與ANS-1相同)

題目：

$$\cos(x+y)dx + (3y^2 + 2y + \cos(x+y))dy = 0 \quad \text{或} \quad y' = \frac{dy}{dx} = \frac{-\cos(x+y)}{3y^2 + 2y + \cos(x+y)}$$

解題步驟：

(1) 確認是否為正合 ODE：

$$\frac{\partial \cos(x+y)}{\partial y} = \frac{\partial (3y^2 + 2y + \cos(x+y))}{\partial x} = -\sin(x+y)$$

(2-1) From $\frac{\partial u}{\partial x} = M = \cos(x+y)$ 求 u

$$\Rightarrow u = \int \cos(x+y)dx + k(y) = \sin(x+y) + k(y) \dots\dots\dots (c)$$

$$(c) \text{對 } y \text{ 微分} \Rightarrow \frac{\partial u}{\partial y} = \frac{\partial}{\partial y}[\sin(x+y)] + \frac{d}{dy}[k(y)] = \cos(x+y) + \frac{d}{dy}[k(y)] = N$$

$$= 3y^2 + 2y + \cos(x+y) \quad (\text{比較等號兩邊項式})$$

$$\Rightarrow \frac{d}{dy}[k(y)] = 3y^2 + 2y \quad (\text{求出 } k(y))$$

$$\Rightarrow k(y) = y^3 + y^2 + c' \quad (\text{帶入 } (c) \text{式})$$

$$\Rightarrow u = \sin(x+y) + y^3 + y^2 + c' = c''$$

$$\Rightarrow u = \sin(x+y) + y^3 + y^2 = c \quad (\text{一定要記得加上 } = c)$$

(2-2) From $\frac{\partial u}{\partial y} = N = 3y^2 + 2y + \cos(x+y)$ 求 u

$$\Rightarrow u = \int 3y^2 + 2y + \cos(x+y)dy + l(x) = y^3 + y^2 + \sin(x+y) + l(x) \dots\dots (d)$$

$$(d) \text{對 } x \text{微分} \Rightarrow \frac{\partial u}{\partial x} = \frac{\partial}{\partial x}[y^3 + y^2 + \sin(x+y)] + \frac{d}{dx}[l(x)] = \cos(x+y) + \frac{d}{dx}[l(x)] = M$$

$$= \cos(x+y) \quad (\text{比較等號兩邊項式})$$

$$\Rightarrow \frac{d}{dx}[l(x)] = 0 \quad (\text{求出 } l(x))$$

$$\Rightarrow l(x) = c' \quad (\text{帶入 } (d) \text{式})$$

$$\Rightarrow u = \sin(x+y) + y^3 + y^2 + c' = c''$$

$$\Rightarrow u = \sin(x+y) + y^3 + y^2 = c \quad (\text{一定要記得加上 } =c)$$

(2-1)與(2-2)的計算法，答案一定會一樣！

Step 3. Checking an implicit solution. We can check by differentiating the implicit solution $u(x, y) = c$ implicitly and see whether this leads to the given ODE (7):

$$(9) \quad du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy = \cos(x+y) dx + (\cos(x+y) + 3y^2 + 2y) dy = 0.$$

驗證du等於(7)式

This completes the check.

自行練習

EXAMPLE 2 An Initial Value Problem

Solve the initial value problem

$$(10) \quad (\cos y \sinh x + 1) dx - \sin y \cosh x dy = 0, \quad y(1) = 2.$$

Solution. You may verify that the given ODE is exact. We find u . For a change, let us use (6*),

$$u = -\int \sin y \cosh x dy + l(x) = \cos y \cosh x + l(x).$$

From this, $\partial u / \partial x = \cos y \sinh x + 1 = M$, $\partial u / \partial y = -\sin y \cosh x = N$. Hence $dl/dx = 1$. By integration, $l(x) = x + c^*$. This gives the general solution $u(x, y) = \cos y \cosh x + x = c$. Using the initial condition, $\cos 2 \cosh 1 + 1 = 0.358 = c$. Hence the answer is $\cos y \cosh x + x = 0.358$. Figure 17 shows the particular solutions for $c = 0, 0.358$ (thicker curve), 1, 2, 3. Check that the answer satisfies the ODE. (Proceed as in Example 1.) Also check that the initial condition is satisfied.

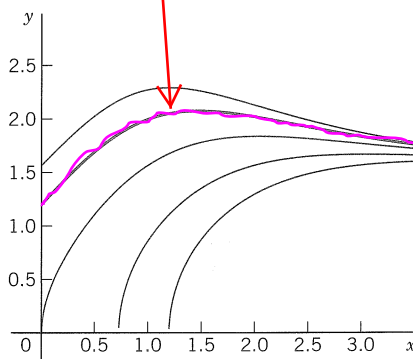


Fig. 17. Particular solutions in Example 2

$$M = -y, N = x \Rightarrow \frac{\partial M}{\partial y} = -1 \neq \frac{\partial N}{\partial x} = 1$$

判斷式不滿足, 非正合ODE

永遠不可能為2x, 因為k(y)只是y的函數

EXAMPLE 3

WARNING! Breakdown in the Case of Nonexactness

The equation $-y dx + x dy = 0$ is not exact because $M = -y$ and $N = x$, so that in (5), $\partial M / \partial y = -1$ but $\partial N / \partial x = 1$. Let us show that in such a case the present method does not work. From (6),

$$u = \int M dx + k(y) = -xy + k(y), \quad \text{hence} \quad \frac{\partial u}{\partial y} = -x + \frac{dk}{dy} = N = x$$

Now, $\partial u / \partial y$ should equal $N = x$, by (4b). However, this is impossible because $k(y)$ can depend only on y . Try (6*); it will also fail. Solve the equation by another method that we have discussed.

$$\frac{\partial P}{\partial y} \neq \frac{\partial Q}{\partial x}$$

$$Pdx + Qdy = 0$$

非正合型式

化簡成

x積分因子

重點：如何找到積分因子呢？

正合型式

$$Mdx + Ndy = 0$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Reduction to Exact Form. Integrating Factors

The ODE in Example 3 is $-y dx + x dy = 0$. It is not exact. However, if we multiply it by $1/x^2$, we get an exact equation [check exactness by (5)!],

$$\frac{-y dx + x dy}{x^2} = -\frac{y}{x^2} dx + \frac{1}{x} dy = d\left(\frac{y}{x}\right) = 0.$$

乘上1/x^2造成正合型式

Integration of (11) then gives the general solution $y/x = c = \text{const.}$

分式微分

$$\left(\frac{g}{f}\right)' = \frac{g'f - gf'}{f^2}$$

This example gives the idea. All we did was to multiply a given nonexact equation, say,

$$(12) \quad P(x, y) dx + Q(x, y) dy = 0, \quad \text{非正合型式}$$

by a function F that, in general, will be a function of both x and y . $\times$ 積分因子 F an equation

$$(13) \quad Mdx + Ndy = FP dx + FQ dy = 0 \quad \text{正合型式}$$

that is exact, so we can solve it as just discussed. Such a function $F(x, y)$ is then called an **integrating factor** of (12).

EXAMPLE 4 Integrating Factor

The integrating factor in (11) is $F = 1/x^2$. Hence in this case the exact equation (13) is

$$FP dx + FQ dy = \frac{-y dx + x dy}{x^2} = d\left(\frac{y}{x}\right) = 0. \quad \text{Solution} \quad \frac{y}{x} = c.$$

都可以當作
積分因子

積分因子有許多個...但要如何
找到呢?

These are straight lines $y = cx$ through the origin. (Note that $x = 0$ is also a solution of $-y dx + x dy = 0$.) It is remarkable that we can readily find other integrating factors for the equation $-y dx + x dy = 0$, namely, $1/y^2$, $1/(xy)$, and $1/(x^2 + y^2)$, because

$$(14) \quad \frac{-y dx + x dy}{y^2} = d\left(\frac{x}{y}\right), \quad \frac{-y dx + x dy}{xy} = -d\left(\ln \frac{x}{y}\right), \quad \frac{-y dx + x dy}{x^2 + y^2} = d\left(\arctan \frac{y}{x}\right).$$

How to Find Integrating Factors

In simpler cases we may find integrating factors by inspection or perhaps after some trials, keeping (14) in mind. In the general case, the idea is the following.

For $M dx + N dy = 0$ the exactness condition (5) is $\partial M / \partial y = \partial N / \partial x$. Hence for (13), $FP dx + FQ dy = 0$, the exactness condition is

$$(15) \quad \frac{\partial}{\partial y} (FP) = \frac{\partial}{\partial x} (FQ).$$

By the product rule, with subscripts denoting partial derivatives, this gives

$$F_y P + FP_y = F_x Q + FQ_x. \rightarrow \text{太複雜, 進行簡化}$$

In the general case, this would be complicated and useless. So we follow the **Golden Rule**: If you cannot solve your problem, try to solve a simpler one—the result may be useful (and may also help you later on). Hence we look for an integrating factor depending only on **one** variable: fortunately, in many practical cases, there are such factors, as we shall see. Thus, let $F = F(x)$. Then $F_y = 0$, and $F_x = F' = dF/dx$, so that (15) becomes

$$FP_y = F'Q + FQ_x.$$

Dividing by FQ and reshuffling terms, we have P, Q 代入求R

$$\frac{1}{F} \frac{dF}{dx} = R, \quad \text{where} \quad R = \frac{1}{Q} \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x} \right)$$

求得積分因子 $F(x)$ 的公式, 相當重要!!

R 僅為 x 的函數, 若是你得到 $R(x, y)$ 不是算錯, 就是不合理。

$$F = \int R dx \quad (\times c) \quad \text{可省略}$$

為什麼沒有 c^* ?

$$\frac{\partial P}{\partial y} \neq \frac{\partial Q}{\partial x}$$

$$Pdx + Qdy = 0 \quad \times F \rightarrow \text{非正合方程式}$$

$F = F(x, y)$ 太複雜, 變成單一變數之函數 $F = F(x)$

$$\frac{1}{Q} \frac{\partial P}{\partial y} = \frac{1}{F} \frac{dF}{dx} + \frac{1}{Q} \frac{\partial Q}{\partial x}$$

$$\Rightarrow \frac{1}{F} \frac{dF}{dx} = \frac{1}{Q} \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x} \right) = R$$

$$\text{故 } \frac{dF}{F} = R dx$$

$$\Rightarrow \int \frac{dF}{F} = \int R dx \Rightarrow \ln F = \int R dx \quad \text{故 } F = \int R dx \quad (\times c) \quad \text{可省略}$$

This proves the following theorem.

$$R(x) = \frac{1}{Q} \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x} \right)$$

$$\frac{\partial P}{\partial y} \neq \frac{\partial Q}{\partial x} \quad \text{THEOREM 1}$$

非正合型式
 $Pdx + Qdy = 0$

↓ x積分因子
 $F(x)$ 或 $F^*(y)$

正合型式
 $F Pdx + F Qdy = 0$

Integrating Factor $F(x)$

If (12) is such that the right side R of (16) depends only on x , then (12) has an integrating factor $F = F(x)$, which is obtained by integrating (16) and taking exponents on both sides. F 是 x 的函數！！

(17)

$$F(x) = \exp \int R(x) dx.$$

Similarly, if $F^* = F^*(y)$, then instead of (16) we get

$$(18) \quad \frac{1}{F^*} \frac{dF^*}{dy} = R^*, \quad \text{where} \quad R^* = \frac{1}{P} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right)$$

and we have the companion

求得積分因子 $F(y)$ 的公式，相當重要！！
注意(16)式與(18)式的差異.....

THEOREM 2

Integrating Factor $F^*(y)$

If (12) is such that the right side R^* of (18) depends only on y , then (12) has an integrating factor $F^* = F^*(y)$ which is obtained from (18) in the form

F 是 y 的函數！！

(19)

$$F^*(y) = \exp \int R^*(y) dy.$$

注意解題程序！！

EXAMPLE 5

Application of Theorems 1 and 2. Initial Value Problem

Using Theorem 1 or 2, find an integrating factor and solve the initial value problem

$$(20) \quad (e^{x+y} + ye^y) dx + (xe^y - 1) dy = 0, \quad y(0) = -1$$

Solution. Step 1. *Nonexactness.* The exactness check fails:

$$\frac{\partial P}{\partial y} = \frac{\partial}{\partial y} (e^{x+y} + ye^y) = e^{x+y} + e^y + ye^y \quad \text{but} \quad \frac{\partial Q}{\partial x} = \frac{\partial}{\partial x} (xe^y - 1) = e^y.$$

Step 2. *Integrating factor. General solution.* Theorem 1 fails because R [the right side of (16)] depends on both x and y .

② 求得 $F(x)$ 或 $F^*(y)$

$$R = \frac{1}{Q} \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x} \right) = \frac{1}{xe^y - 1} (e^{x+y} + e^y + ye^y - e^y) = \frac{e^{x+y} + ye^y}{xe^y - 1}. \quad \text{不合理! Why?}$$

Try Theorem 2. The right side of (18) is

$$R^* = \frac{1}{P} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) = \frac{1}{e^{x+y} + ye^y} (e^y - e^{x+y} - e^y - ye^y) = -1.$$

Hence (19) gives the integrating factor $F^*(y) = e^{-y}$. From this result and (20) you get the exact equation

$$(e^x + y) dx + (x - e^{-y}) dy = 0.$$

正合型式
 $F^* P dx + F^* Q dy = 0$

之後解題方法與步驟如前正合ODE!!

$$\frac{\partial u}{\partial x} = M \rightarrow u = \int M dx + k(y)$$

Test for exactness; you will get 1 on both sides of the exactness condition. By integration, using (4a),

$$u = \int (e^x + y) dx = e^x + xy + k(y). = c^{**}$$

Differentiate this with respect to y and use (4b) to get

$$\frac{\partial u}{\partial y} = N \rightarrow \frac{\partial u}{\partial y} = x + \frac{dk}{dy} = N = x - e^{-y} \quad \frac{dk}{dy} = -e^{-y}, \quad k = e^{-y} + c^*.$$

Hence the general solution is

比較項式

通解

$$u(x, y) = e^x + xy + e^{-y} = c. = c^{**} - c^* = c$$

利用初始條件，求特解！

Step 3. Particular solution. The initial condition $y(0) = -1$ gives $u(0, -1) = 1 + 0 + e = 3.72$. Hence the answer is $e^x + xy + e^{-y} = 1 + e = 3.72$. Figure 18 shows several particular solutions obtained as level curves of $u(x, y) = c$, obtained by **特解** convenient way in cases in which it is impossible or difficult to cast a solution into explicit form. Note the curve that (nearly) satisfies the initial condition.

Step 4. Checking. Check by substitution that the answer satisfies the given equation as well as the initial condition.

即 $x=0, y=-1$

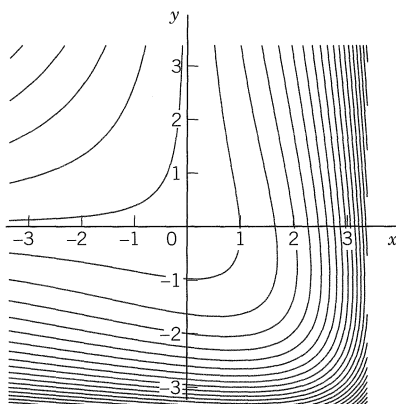


Fig. 18. Particular solutions in Example 5

PROBLEM SET 1.4

1-14 ODEs. INTEGRATING FACTORS

Test for exactness. If exact, solve. If not, use an integrating factor as given or obtained by inspection or by the theorems in the text. Also, if an initial condition is given, find the corresponding particular solution.

1. $2xy dx + x^2 dy = 0$
2. $x^3 dx + y^3 dy = 0$
3. $\sin x \cos y dx + \cos x \sin y dy = 0$
4. $e^{3\theta}(dr + 3r d\theta) = 0$
5. $(x^2 + y^2) dx - 2xy dy = 0$
6. $3(y + 1) dx = 2x dy, \quad (y + 1)x^{-4}$
7. $2x \tan y dx + \sec^2 y dy = 0$

多做習題
增加解題
熟練度！

8. $e^x(\cos y dx - \sin y dy) = 0$
9. $e^{2x}(2 \cos y dx - \sin y dy) = 0, \quad y(0) = 0$
10. $y dx + [y + \tan(x + y)] dy = 0, \quad \cos(x + y)$
11. $2 \cosh x \cos y dx = \sinh x \sin y dy$
12. $(2xy dx + dy)e^{x^2} = 0, \quad y(0) = 2$
13. $e^{-y} dx + e^{-x}(-e^{-y} + 1) dy = 0, \quad F = e^{x+y}$
14. $(a + 1)y dx + (b + 1)x dy = 0, \quad y(1) = 1, \quad F = x^a y^b$

15. **Exactness.** Under what conditions for the constants a, b, k, l is $(ax + by) dx + (kx + ly) dy = 0$ exact? Solve the exact ODE.

題目 $P(x, y)dx + Q(x, y)dy = 0$

解題步驟：

(1) 確認是否為正合ODE： $\frac{\partial P}{\partial y} \neq \frac{\partial Q}{\partial x}$ (非正合ODE)

(2) 求乘積分因子 $F(x)$ 或 $F(y)$

(證-1) $F(x)$ 的公式來源

$$FPdx + FQdy = 0$$

$$\frac{\partial FP}{\partial y} = \frac{\partial FQ}{\partial x} \Rightarrow F_y P + FP_y = F_x Q + FQ_x \quad (F(x, y) \text{太複雜進行化簡})$$

$$\text{令 } F \text{ 為單一變數 } F(x) \Rightarrow F_y = 0 \Rightarrow FP_y = F_x Q + FQ_x$$

$$\text{等號兩邊同除以 } FQ \Rightarrow \frac{P_y}{Q} = \frac{F_x}{F} + \frac{Q_x}{Q} \Rightarrow \frac{1}{Q} \frac{\partial P}{\partial y} = \frac{1}{F} \frac{dF}{dx} + \frac{1}{Q} \frac{\partial Q}{\partial x}$$

$$\text{分離變數法求解 } F(x) \Rightarrow \frac{1}{F} \frac{dF}{dx} = \frac{1}{Q} \frac{\partial P}{\partial y} - \frac{1}{Q} \frac{\partial Q}{\partial x} = \frac{1}{Q} \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x} \right) = R(x)$$

$$\Rightarrow \frac{dF}{F} = Rdx \Rightarrow \int \frac{dF}{F} = \int Rdx$$

$$\Rightarrow \ln F = \int Rdx \Rightarrow F = e^{\int Rdx}$$

(證-2) $F(y)$ 的公式來源 (同理可證)

$$FPdx + FQdy = 0$$

$$\frac{\partial FP}{\partial y} = \frac{\partial FQ}{\partial x} \Rightarrow F_y P + FP_y = F_x Q + FQ_x \quad (F(x, y) \text{太複雜進行化簡})$$

$$\text{令 } F \text{ 為單一變數 } F(y) \Rightarrow F_x = 0 \Rightarrow F_y P + FP_y = FQ_x$$

$$\text{等號兩邊同除以 } FP \Rightarrow \frac{Q_x}{P} = \frac{F_y}{F} + \frac{P_y}{P} \Rightarrow \frac{1}{P} \frac{\partial Q}{\partial x} = \frac{1}{F} \frac{dF}{dy} + \frac{1}{P} \frac{\partial P}{\partial y}$$

$$\text{分離變數法求解 } F(y) \Rightarrow \frac{1}{F} \frac{dF}{dy} = \frac{1}{P} \frac{\partial Q}{\partial x} - \frac{1}{P} \frac{\partial P}{\partial y} = \frac{1}{P} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) = R^*(y)$$

$$\Rightarrow \frac{dF}{F} = R^* dy \Rightarrow \int \frac{dF}{F} = \int R^* dy$$

$$\Rightarrow \ln F = \int R^* dy \Rightarrow F = e^{\int R^* dy}$$

當 $F(x)$ 或 $F(y)$ 求出，先驗算是否滿足 $\frac{\partial FP}{\partial y} = \frac{\partial FQ}{\partial x}$ (正合方程式條件)

若 $\frac{\partial FP}{\partial y} \neq \frac{\partial FQ}{\partial x}$ 代表 $F(x)$ 或 $F(y)$ 求錯了，趕快再檢查重新計算。

若 $\frac{\partial FP}{\partial y} = \frac{\partial FQ}{\partial x}$ 代表已獲得正合方程式，利用正合方程式的解題步驟求解。

題目：非正合問題

$$(e^{x+y} + ye^y)dx + (xe^y - 1)dy = 0 \quad \text{或} \quad y' = \frac{dy}{dx} = \frac{-(e^{x+y} + ye^y)}{(xe^y - 1)}, \quad y(0) = -1$$

即 $P(x, y)dx + Q(x, y)dy = 0$

解題步驟：

(1) 確認是否為正合ODE

$$\left. \begin{aligned} \frac{\partial P}{\partial y} &= \frac{\partial(e^{x+y} + ye^y)}{\partial y} = e^{x+y} + e^y + ye^y \\ \frac{\partial Q}{\partial x} &= \frac{\partial(xe^y - 1)}{\partial x} = e^y \end{aligned} \right\} \quad \frac{\partial P}{\partial y} \neq \frac{\partial Q}{\partial x} \quad (\text{非正合})$$

(2) 求積分因子 $F(x)$ 或 $F(y)$

$$R(x) = \frac{1}{Q} \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x} \right) = \frac{1}{xe^y - 1} (e^{x+y} + e^y + ye^y - e^y) \quad \text{不合理，因} R \text{僅能為} x \text{的函數}$$

故 $F(x)$ 無法求出

$$R^*(y) = \frac{1}{P} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) = \frac{1}{e^{x+y} + ye^y} (e^y - e^{x+y} - e^y - ye^y) = -1$$

$$F(y) = e^{\int R^*(y) dy} = e^{-y}$$

(4) 求解正合方程式

$$\Rightarrow u = \int (e^x + y)dx + k(y) = (e^x + xy) + k(y) = C^{**} \dots\dots (a)$$

$$(a) \text{對} y \text{微分} \Rightarrow \frac{\partial u}{\partial y} = \frac{\partial}{\partial y} (e^x + xy) + \frac{d}{dy} [k(y)] = x + \frac{d}{dy} [k(y)] = N = x - e^{-y}$$

(比較等號兩邊項式)

$$\Rightarrow \frac{d}{dy} [k(y)] = -e^{-y} \quad (\text{求出} k(y))$$

$$\Rightarrow k(y) = e^{-y} + C^* \quad (\text{帶入}(a)\text{式})$$

$$\Rightarrow u = (e^x + xy) + e^{-y} + C^* = C^{**}$$

$$\Rightarrow u = (e^x + xy) + e^{-y} = C \quad (\text{一定要記得加上} = C)$$

$$\text{初始條件 } y(0) = -1 \Rightarrow u(0, -1) = 1 + 0 + e = 3.72 = C$$

$$\text{特解 } u = (e^x + xy) + e^{-y} = 3.72$$

16. TEAM PROJECT. Solution by Several Methods.

Show this as indicated. Compare the amount of work.

(a) $e^y(\sinh x \, dx + \cosh x \, dy) = 0$ as an exact ODE and by separation.

(b) $(1 + 2x) \cos y \, dx + dy/\cos y = 0$ by Theorem 2 and by separation.

(c) $(x^2 + y^2) \, dx - 2xy \, dy = 0$ by Theorem 1 or 2 and by separation with $v = y/x$.

(d) $3x^2 y \, dx + 4x^3 \, dy = 0$ by Theorems 1 and 2 and by separation.

(e) Search the text and the problems for further ODEs that can be solved by more than one of the methods discussed so far. Make a list of these ODEs. Find further cases of your own.

17. WRITING PROJECT. Working Backward.

Working backward from the solution to the problem is useful in many areas. Euler, Lagrange, and other great masters did it. To get additional insight into the idea of integrating factors, start from a $u(x, y)$ of your choice, find $du = 0$, destroy exactness by

18. CAS PROJECT. Graphing Particular Solutions.

Graph particular solutions of the following ODE, proceeding as explained.

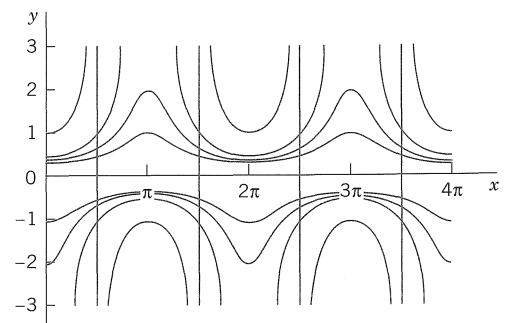
$$(21) \quad dy - y^2 \sin x \, dx = 0.$$

(a) Show that (21) is not exact. Find an integrating factor using either Theorem 1 or 2. Solve (21).

(b) Solve (21) by separating variables. Is this simpler than (a)?

(c) Graph the seven particular solutions satisfying the following initial conditions $y(0) = 1$, $y(\pi/2) = \pm \frac{1}{2}$, $\pm \frac{2}{3}$, ± 1 (see figure below).

(d) Which solution of (21) do we not get in (a) or (b)?



Particular solutions in CAS Project 18

學習重點：

1. 如何求解線性 ODEs 的問題？

→ 分為齊次問題與非齊次問題

2. 如何從非線性 ODEs 轉換成線性 ODEs？

白努利方程式

1.5 Linear ODEs. Bernoulli Equation.

Population Dynamics

人口動態(變化)學

Linear ODEs or ODEs that can be transformed to linear form are models of various phenomena, for instance, in physics, biology, population dynamics, and ecology, as we shall see. A first-order ODE is said to be **linear** if it can be brought into the form

(1)

$$y' + p(x)y = r(x)$$

標準型式, y' 的係數必定為 1

by algebra, and **nonlinear** if it cannot be brought into this form.

The defining feature of the linear ODE (1) is that it is linear in both the unknown function y and its derivative $y' = dy/dx$, whereas p and r may be **any** given functions of x . If in an application the independent variable is time, we write t instead of x .

If the first term is $f(x)y'$ (instead of y'), divide the equation by $f(x)$ to get the **standard form** (1), with y' as the first term, which is practical.

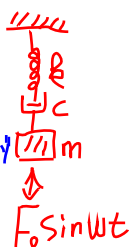
For instance, $y' \cos x + y \sin x = x$ is a linear ODE, and its standard form is $y' + y \tan x = x \sec x$.

The function $r(x)$ on the right may be a force, and the solution $y(x)$ a displacement in a motion or an electrical current or some other physical quantity. In engineering, $r(x)$ is frequently called the **input**, and $y(x)$ is called the **output** or the **response** to the input (and, if given, to the initial condition).

解題關鍵, 先化成標準型式, 如

$$f(x)y' + p(x)y = r(x)$$

$$\Rightarrow y' + \frac{p(x)}{f(x)}y = \frac{r(x)}{f(x)}$$



$$m y'' + c y' + k y = F_0 \sin \omega t$$

$$\Rightarrow y'' + \frac{c}{m} y' + \frac{k}{m} y = \frac{F_0}{m} \sin \omega t$$

input

$y(x)$

響應

齊次的 ($r(x)=0$)

非正合型式

Homogeneous Linear ODE. We want to solve (1) in some interval $a < x < b$, call it J , and we begin with the simpler special case that $r(x)$ is zero for all x in J . (This is sometimes written $r(x) \equiv 0$.) Then the ODE (1) becomes

(2)

$$\frac{dy}{dx} + p(x)y = 0 \iff y' = -p(x)y \Rightarrow y = ce^{\int p(x) dx}$$

利用分離變數法求解

and is called **homogeneous**. By separating variables and integrating we then obtain

$$\int \frac{dy}{y} = \int -p(x) dx + c^* \quad \text{thus} \quad \ln |y| = -\int p(x) dx + c^*.$$

Taking exponents on both sides, we obtain the general solution of the homogeneous ODE (2),

$$y(x) = ce^{-\int p(x) dx}$$

$$(c = \pm e^{c^*} \text{ when } y \geq 0);$$

不建議背公式

here we may also choose $c = 0$ and obtain the **trivial solution** $y(x) = 0$ for all x in that interval.

恆零解

非齊次的 ($r(x) \neq 0$)

Nonhomogeneous Linear ODE. We now solve (1) in the case that $r(x)$ in (1) is not everywhere zero in the interval J considered. Then the ODE (1) is called **nonhomogeneous**. It turns out that in this case, (1) has a pleasant property; namely, it has an integrating factor depending only on x . We can find this factor $F(x)$ by Theorem 1 in the previous section or we can proceed directly, as follows. We multiply (1) by $F(x)$, obtaining

等號兩邊 $\times F$

$$(1^*) \quad (Fy)' + pFy = rF.$$

(視為積分因子)

The left side is the derivative $(Fy)' = F'y + Fy'$ of the product Fy if

$$pF = F', \quad \text{thus} \quad pF = F' = \frac{dF}{dx}$$

By separating variables, $dF/F = p dx$. By integration, writing $h = \int p dx$,

$$\ln |F| = h = \int p dx, \quad \text{thus} \quad F = e^h.$$

(F可視為積分因子, 以非正合ODE求解)

$$h' = p$$

代入(1*)式中

With this F and $h' = p$, Eq. (1*) becomes

$$e^h y' + h' e^h y = e^h y' + (e^h)' y = (e^h y)' = r e^h$$

By integration,

$$(e^h y)' = \int e^h r dx + c.$$

等號兩邊取積分

Dividing by e^h , we obtain the desired solution formula

(1)式(一定要先化成標準式)的通解

$$y(x) = e^{-h} \left(\int e^h r dx + c \right), \quad h = \int p(x) dx.$$

公式推導與記憶, 重要!!
(注意括號的位置)

$$\frac{r(x)}{f(x)}$$

$$\frac{p(x)}{f(x)}$$

在(1)式的 y' 項係數為1的情況下, 代入的 $r(x)$ 與 $p(x)$ 才是正確, 不要被騙了!!

標準型式, y' 的係數必定為1
看到題目先化成標準型式, 如

$$f(x)y' + p(x)y = r(x)$$

$$\Rightarrow y' + \frac{p(x)}{f(x)}y = \frac{r(x)}{f(x)}$$

求積分因子

$$(1) \text{式 } y' + p y = r$$

標準型式

$$(AB)' = A'B + AB'$$

$$(e^{ax})' = a e^{ax}$$

$$(\int p dx)' = p$$

This reduces solving (1) to the generally simpler task of evaluating integrals. For ODEs for which this is still difficult, you may have to use a numeric method for integrals from Sec. 19.5 or for the ODE itself from Sec. 21.1. We mention that h has nothing to do with $h(x)$ in Sec. 1.1 and that the constant of integration in h does not matter; see Prob. 2.

The structure of (4) is interesting. The only quantity depending on a given initial condition is c . Accordingly, writing (4) as a sum of two terms,

$$(4^*) \quad y(x) = e^{-h} \int e^h r dx + c e^{-h},$$

we see the following:

r 相關的特解

齊次解, 即(3)式
(與系統的特性有關)

$$(5) \quad \text{Total Output} = \text{Response to the Input } r + \text{Response to the Initial Data.}$$

EXAMPLE 1

First-Order ODE, General Solution, Initial Value Problem

Solve the initial value problem

注意! 是否為標準形式

$$y' + y \tan x = \sin 2x, \quad y(0) = 1.$$

Solution. Here $p = \tan x$, $r = \sin 2x = 2 \sin x \cos x$, and

先找出 p, r

$$h = \int p dx = \int \tan x dx = \ln |\sec x|.$$

② 求出 h

From this we see that in (4),

$e^h = \sec x$,
積分因子

$$e^{-h} = \cos x,$$

$$\int \frac{\sin x}{\cos x} dx = -\int \frac{-\sin x}{\cos x} dx = -\ln |\cos x|$$

$$e^h r = (\sec x)(2 \sin x \cos x) = 2 \sin x, \quad ③$$

and the general solution of our equation is

$$④ \quad y(x) = \cos x \left(2 \int \sin x dx + c \right) = c \cos x - 2 \cos^2 x. \quad \text{通解!}$$

From this and the initial condition, $1 = c \cdot 1 - 2 \cdot 1^2$; thus $c = 3$ and the solution of our initial value problem is $y = 3 \cos x - 2 \cos^2 x$. Here $3 \cos x$ is the response to the initial data, and $-2 \cos^2 x$ is the response to the input $\sin 2x$. 特解!

輸入 $\sin 2x$
的響應

EXAMPLE 2

Electric Circuit

Model the **RL-circuit** in Fig. 19 and solve the resulting ODE for the current $I(t)$ A (amperes), where t is time. Assume that the circuit contains as an **EMF** $E(t)$ (electromotive force) a battery of $E = 48$ V (volts), which is constant, a **resistor** of $R = 11 \Omega$ (ohms), and an **inductor** of $L = 0.1$ H (henrys), and that the current is initially zero.

初始條件 $I(0) = 0$

Physical Laws. A current I in the circuit causes a **voltage drop** RI across the resistor (**Ohm's law**) and a voltage drop $LI' = L dI/dt$ across the conductor, and the sum of these two voltage drops equals the EMF (**Kirchhoff's Voltage Law, KVL**).

Remark. In general, KVL states that "The voltage (the electromotive force EMF) impressed on a closed loop is equal to the sum of the voltage drops across all the other elements of the loop." For Kirchhoff's Current Law (KCL) and historical information, see footnote 7 in Sec. 2.9.

Solution. According to these laws the model of the **RL-circuit** is $LI' + RI = E(t)$, in standard form

$$(6) \quad I' + \frac{R}{L} I = \frac{E(t)}{L}.$$

① 先化成標準形式
(I' 的係數必定為 1)

$$E(t) - RI - LI' = 0$$

係數一定要化為 1

$$y' + p(x)y = r(x)$$

$$h = \int p(x) dx$$

見非正合 ODE 法
求解補充!!

$$\ln |f(x)| = \int \frac{f'(x)}{f(x)} dx$$

初始值的
響應

【利用非正合 ODE 法求解】

To solve $y' + y \tan x = \sin 2x$

【Solution】

$$y' + y \tan x = \sin 2x$$

先改成正合 ODE 的形式

$$\Rightarrow \frac{dy}{dx} + (y \tan x - \sin 2x) = 0$$

$$\Rightarrow dy + (y \tan x - \sin 2x)dx = 0 \Rightarrow \underline{(y \tan x - \sin 2x)dx + dy = 0} \Rightarrow P = y \tan x - \sin 2x, Q = 1$$

利用公式求積分因子 $\frac{1}{F} \frac{dF}{dx} = R, R = \frac{1}{Q} \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x} \right)$

$$R = \frac{1}{1} \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x} \right) = \tan x \Rightarrow \ln F = \int \tan x dx = \int \frac{\sin x}{\cos x} dx = \ln \frac{1}{\cos x} = \ln \sec x$$

$$\Rightarrow F = e^{\ln \sec x} = \sec x = \frac{1}{\cos x} \quad (\text{積分因子}) \quad \text{"} - \int \frac{\sin x}{\cos x} dx = -\ln \cos x$$

$$F(y \tan x - \sin 2x)dx + Fdy = 0 \Rightarrow \frac{1}{\cos x} (y \frac{\sin x}{\cos x} - 2 \sin x \cos x)dx + \frac{1}{\cos x} dy = 0$$

$$\Rightarrow (y \frac{\sin x}{\cos^2 x} - 2 \sin x)dx + \frac{1}{\cos x} dy = Mdx + Ndy = 0$$

驗證是否為正合 ODE

$$\frac{\partial M}{\partial y} = \frac{\sin x}{\cos^2 x} = \frac{\partial N}{\partial x} = \frac{\partial}{\partial x} \left(\frac{1}{\cos x} \right) = \frac{\sin x}{\cos^2 x} \quad (\text{滿足正合 ODE})$$

求解

$$\begin{aligned} u = \int Mdx + k(y) &= \int (y \frac{\sin x}{\cos^2 x} - 2 \sin x)dx + k(y) = y \int \frac{\sin x}{\cos^2 x} dx - 2 \int \sin x dx + k(y) = y \int \frac{-d \cos x}{\cos^2 x} + 2 \cos x + k(y) \\ &= y \cos^{-1} x + 2 \cos x + k(y) = c' \dots\dots\dots(A) \end{aligned}$$

求 $k(y)$

$$\frac{\partial u}{\partial y} = \cos^{-1} x + \frac{dk(y)}{dy} = N = \frac{1}{\cos x} \Rightarrow \frac{dk(y)}{dy} = 0 \Rightarrow k(y) = c'' \text{ 帶入 (A) 式中}$$

求 y 解

$$\Rightarrow u = y \cos^{-1} x + 2 \cos x + c'' = c'$$

$$\Rightarrow u = \underline{y \cos^{-1} x + 2 \cos x = c}$$

$$\text{等號兩邊乘上 } \cos x \Rightarrow y = c \cos x - 2 \cos^2 x \dots\dots\dots \text{ANS}$$

$$\begin{aligned}
 h &= \int P(t) dt \\
 &= \int \frac{R}{L} dt \\
 &= \frac{R}{L} t
 \end{aligned}$$

We can solve this linear ODE by (4) with $x = t$, $y = I$, $p = R/L$, $h = (R/L)t$, obtaining the general solution

$$I = e^{-(R/L)t} \left(\int e^{(R/L)t} \frac{E(t)}{L} dt + c \right).$$

By integration,

$$(7) \quad I = e^{-(R/L)t} \left(\frac{E}{L} \frac{e^{(R/L)t}}{R/L} + c \right) = \frac{E}{R} + ce^{-(R/L)t}.$$

In our case, $R/L = 11/0.1 = 110$ and $E(t) = 48/0.1 = 480 = \text{const}$; thus,

$$I = \frac{48}{11} + ce^{-110t}.$$

In modeling, one often gets better insight into the nature of a solution (and smaller roundoff errors) by inserting given numeric data only near the end. Here, the general solution (7) shows that the current approaches the limit $E/R = 48/11$ the larger R/L is, in our case, $R/L = 11/0.1 = 110$, and the approach is very fast, from below if $I(0) < 48/11$ or from above if $I(0) > 48/11$. If $I(0) = 48/11$, the solution is constant ($48/11$ A). See Fig. 19.

The initial value $I(0) = 0$ gives $I(0) = E/R + c = 0$, $c = -E/R$ and the particular solution

$$(8) \quad I = \frac{E}{R} (1 - e^{-(R/L)t}), \quad \text{thus} \quad I = \frac{48}{11} (1 - e^{-110t}).$$

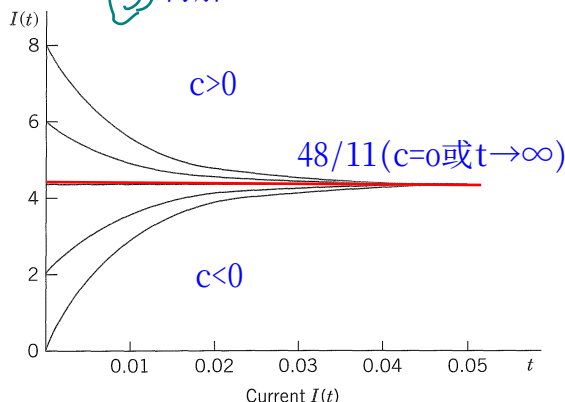
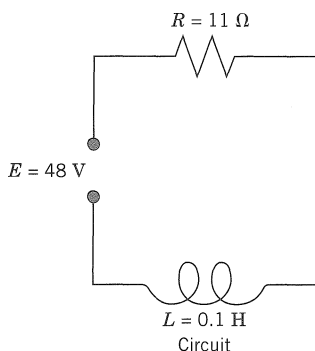


Fig. 19. RL-circuit

柯西荷夫定律

$$\begin{aligned}
 E_0 \sin \omega t - RI - \frac{1}{C} \int I dt \\
 -L I' = 0
 \end{aligned}$$

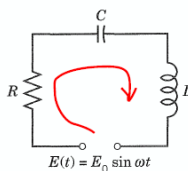


Fig. 61. RLC-circuit

Name	Symbol	Notation	Unit	Voltage Drop
Ohm's Resistor		R Ohm's Resistance	ohms (Ω)	RI
Inductor		L Inductance	henrys (H)	$L \frac{dI}{dt} = \mathcal{L} I'$
Capacitor		C Capacitance	farads (F)	Q/C

Fig. 62. Elements in an RLC-circuit

$$\frac{1}{C} \int I dt$$

$$\begin{aligned}
 y(t) &= e^{-Kt} \int e^{Kt} (A + B \cos \omega t) dt + ce^{-Kt} \\
 &= e^{-Kt} e^{Kt} \left[\frac{A}{K} + \frac{B}{K^2 + \omega^2} (K \cos \omega t + \omega \sin \omega t) \right] + ce^{-Kt} \\
 &= \frac{A}{K} + \frac{B}{K^2 + (\pi/12)^2} \left(K \cos \frac{\pi t}{12} + \frac{\pi}{12} \sin \frac{\pi t}{12} \right) + ce^{-Kt}.
 \end{aligned}$$

The last term decreases to 0 as t increases, practically after a short time and regardless of c (that is, of the initial condition). The other part of $y(t)$ is called the **steady-state solution** because it consists of constant and periodic terms. The entire solution is called the **transient-state solution** because it models the transition from rest to the steady state. These terms are used quite generally for physical and other systems whose behavior depends on time.

Step 3. Particular solution. Setting $t = 0$ in $y(t)$ and choosing $y_0 = 0$, we have

$$y(0) = \frac{A}{K} + \frac{B}{K^2 + (\pi/12)^2} \frac{\pi}{12} K + c = 0, \quad \text{thus} \quad c = -\frac{A}{K} - \frac{KB}{K^2 + (\pi/12)^2}.$$

Inserting this result into $y(t)$, we obtain the particular solution

$$y_{\text{part}}(t) = \frac{A}{K} + \frac{B}{K^2 + (\pi/12)^2} \left(K \cos \frac{\pi t}{12} + \frac{\pi}{12} \sin \frac{\pi t}{12} \right) - \left(\frac{A}{K} + \frac{KB}{K^2 + (\pi/12)^2} \right) e^{-Kt}$$

with the steady-state part as before. To plot y_{part} we must specify values for the constants, say, $A = B = 1$ and $K = 0.05$. Figure 20 shows this solution. Notice that the transition period is relatively short (although K is small), and the curve soon looks sinusoidal; this is the response to the input $A + B \cos(\frac{1}{12} \pi t) = 1 + \cos(\frac{1}{12} \pi t)$.

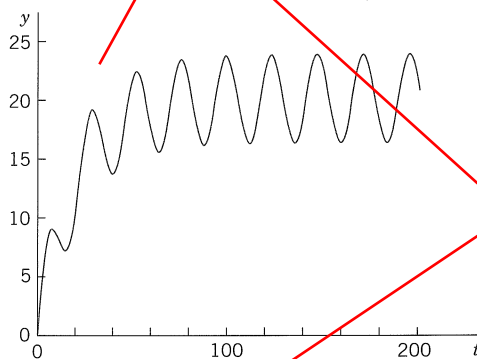


Fig. 20. Particular solution in Example 3

白努利方程式

Reduction to Linear Form. Bernoulli Equation

Numerous applications can be modeled by ODEs that are nonlinear but can be transformed to linear ODEs. One of the most useful ones of these is the **Bernoulli equation**⁷

(9)

$$y' + p(x)y = g(x)y^a$$

(a any real number).

$a=0$ 或 1 , (9)式為線性, 否則為非線性

⁷JAKOB BERNOULLI (1654–1705), Swiss mathematician, professor at Basel, also known for his contribution to elasticity theory and mathematical probability. The method for solving Bernoulli's equation was discovered by Leibniz in 1696. Jakob Bernoulli's students included his nephew NIKLAUS BERNOULLI (1687–1759), who contributed to probability theory and infinite series, and his youngest brother JOHANN BERNOULLI (1667–1748), who had profound influence on the development of calculus, became Jakob's successor at Basel, and had among his students GABRIEL CRAMER (see Sec. 7.7) and LEONHARD EULER (see Sec. 2.5). His son DANIEL BERNOULLI (1700–1782) is known for his basic work in fluid flow and the kinetic theory of gases.

流體力學白努利

學習重點：

非線性ODEs

轉換

線性ODEs

非齊次求解
(以前學過的技巧)

per year. If at some instant the lake has pollution concentration $p = 0.04\%$, how long, approximately, will it take to decrease it to $p/2$, assuming that the inflow is much cleaner, say, it has pollution concentration $p/4$, and the mixture is uniform (an assumption that is only imperfectly true)? First guess.

- 36. Harvesting renewable resources. Fishing.** Suppose that the population $y(t)$ of a certain kind of fish is given by the logistic equation (11), and fish are caught at a rate Hy proportional to y . Solve this so-called *Schaefer model*. Find the equilibrium solutions y_1 and $y_2 (> 0)$ when $H < A$. The expression $Y = Hy_2$ is called the **equilibrium harvest** or **sustainable yield** corresponding to H . Why?
- 37. Harvesting.** In Prob. 36 find and graph the solution satisfying $y(0) = 2$ when (for simplicity) $A = B = 1$ and $H = 0.2$. What is the limit? What does it mean? What if there were no fishing?
- 38. Intermittent harvesting.** In Prob. 36 assume that you fish for 3 years, then fishing is banned for the next 3 years. Thereafter you start again. And so on. This is called *intermittent harvesting*. Describe qualitatively how the population will develop if intermitting is continued periodically. Find and graph the solution for the first 9 years, assuming that $A = B = 1$, $H = 0.2$, and $y(0) = 2$.

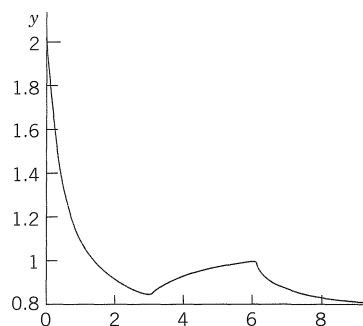


Fig. 23. Fish population in Problem 38

- 39. Extinction vs. unlimited growth.** If in a population $y(t)$ the death rate is proportional to the population, and the birth rate is proportional to the chance encounters of meeting mates for reproduction, what will the model be? Without solving, find out what will eventually happen to a small initial population. To a large one. Then solve the model.
- 40. Air circulation.** In a room containing $20,000 \text{ ft}^3$ of air, 600 ft^3 of fresh air flows in per minute, and the mixture (made practically uniform by circulating fans) is exhausted at a rate of 600 cubic feet per minute (cfm). What is the amount of fresh air $y(t)$ at any time if $y(0) = 0$? After what time will 90% of the air be fresh?

1.6 Orthogonal Trajectories. Optional

An important type of problem in physics or geometry is to find a family of curves that intersects a given family of curves at right angles. The new curves are called **orthogonal trajectories** of the given curves (and conversely). Examples are curves of equal temperature (**isotherms**) and curves of heat flow, curves of equal altitude (**contour lines**) on a map and curves of steepest descent on that map, curves of equal potential (**equipotential curves**, curves of equal voltage—the ellipses in Fig. 24) and curves of electric force (the parabolas in Fig. 24).

Here the **angle of intersection** between two curves is defined to be the angle between the tangents of the curves at the intersection point. *Orthogonal* is another word for *perpendicular*.

In many cases orthogonal trajectories can be found using ODEs. In general, if we consider $G(x, y, c) = 0$ to be a given family of curves in the xy -plane, then each value of c gives a particular curve. Since c is one parameter, such a family is called a **one-parameter family of curves**.

In detail, let us explain this method by a family of ellipses

$$(1) \qquad \frac{1}{2}x^2 + y^2 = c \qquad (c > 0)$$